The Physics of Artificial Intelligence

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Contents

Preface

Actually, I am not an AI or CS student. When the AI explosion occurred 3 years ago, I did not care about it so much. I thought it was a scam; created by the billionaires, who want to take control of the new world. I was not interested in AI in the first place. Everything was nonsense and meaningless, why did I have to care about it?

Like every student living in AI-era, I was forced to learn a subject called "Fundamentals of Artificial Intelligence", 3 credits, in my third year. When the midterm came, I realized that I did not know anything; and the risk of failing this subject was real. 2 hours before midterm, I came to the library and tried to learn something that could save my grade above 4 points. To be honest, the feeling of just studying to merely pass a stupid test and then being done with it was so frustrating. I was watching many tutorial videos on YouTube, and thought "OK, the game is over. I am done.". My brain could not consume any knowledge from them. I was a failure.

Instead of learning like crazy in the last 2 hours left, I just skimmed all the stuffs for 1 hour, and decided to give up. Fortunately, I did misclick a video; this moment would change my vision forever.

"Geoffrey Hinton, Nobel Prize in Physics 2024: Banquet speech"

Wait, did something go wrong here? Nobel Prize in Physics for Artificial Intelligence? My brain turned on because I deeply cared about Physics. "Why did something bullshit receive a Nobel Prize, even in Physics?". I did some searching, and then I found some familiar keywords, such as "Boltzmann Statistics", "Spin-Glass", "Ising Model". "Perhaps I could do some Physics here.", I thought. Time flies so fast, half a year has already passed. I am still confused, but I think I have got an idea. After my first book "Thinking like a Frequentist" was published online, why not try writing about the "bridge" between Physics and AI? At least so far, I have not seen anyone write a sufficiently understandable document on this topic. But the problem is, I am not even a Physics student. Up to now, my Physics knowledge is limited to Halliday textbooks, and some experience with digital and analog electronics. To fill these gaps, I have to learn everything from scratch.

That is why I wrote this book as a concise self-study guide for anyone interested in Physics and AI. Unlike my first book, which was written when my knowledge of Probability and Statistics was already solid; this book was written during my self-study process. I may have made many mistakes or misinterpretations; but I hope you, my readers, can point them out and send feedbacks via email `tinvu1309@gmail.com`. I would be extremely happy if someone could discuss these issues with me.

The structure of this book contains 2 main parts: "Introduction to Statistical Physics" from Chapter 0 to Chapter 7, and "The Physics of Artificial Intelligence" from Chapter 8 to Chapter 15. Although I will try to keep everything as simple as possible; but before reading, you should be familiar with elementary university mathematics concepts, include: *Calculus*, *Linear Algebra* and *Probability and Statistics*. The better your math skills, the higher your physics ability. In physics, you need to understand some basic concepts of *Mechanics and Thermodynamics*; *Electromagnetism* and some applied physics subjects such as *Signals and Systems*, *Control Theory* and *Digital Signal Processing*. Some useful books and lecture notes are listed below:

1. Calculus: Calculus 7E, James Stewart
2. Linear Algebra: Elementary Linear Algebra, Larson - Edwards - Falvo
3. Probability and Statistics: Probability and Statistics for Engineers and Scientists, Walpole - Myers or Thinking like a Frequentist, Tin Vu
4. Mechanics and Thermodynamics, Electromagnetism: Fundamentals of Physics, Halliday and Resnick
5. Signals and Systems: Signals and Systems, Alan V. Oppenheim or my lecture note (Vietnamese)
6. Digital Signals Processing: My lecture note (Vietnamese)
7. Control Theory: Control Systems Engineering, Norman S. Nise or my lecture note (English)

I highly suggest that you should read the first part **without any AI tools** to build physics intuition (I did that too); before diving into the second part of this book, where we "wire" our intuition up with the aid of AI tools. Everything will turn out great as long as we keep trying.

Part I

Introduction to Statistical Physics

Chapter 0

The Zeroth Law of Thermodynamics

0.1 What is Temperature?

If you touch an ice cube, you will feel cold; and if you touch boiling water, you will feel hot. Now you can conclude that ice cube is at low temperature, and boiling water is at high temperature. In common thinking, **temperature** is a quantity that used to measure hotness and coldness of objects. Instead of trusting our feeling, we often use a measuring device called **thermometer** to estimate the temperature.

Example 0.1.1. Using thermometer to measure the temperature of a cup of hot water.

After adjusting the thermometer to normal room temperature, immerse it into a cup; and wait until the mercury inside no longer rises. Now read the temperature displayed on the thermometer.
The waiting time is called **relaxation time**; the state when both thermometer and hot water sharing the same temperature is called **thermal equilibrium**.

0.2 The Zeroth Law of Thermodynamics

Theorem 0.2.1 (The Zeroth Law of Thermodynamics)**.** *If object A and B are each in thermal equilibrium with a third object T, then A and B are in* ***thermal equilibrium*** *with each other.*

In spite of its simplicity, this law provides an independent definition of temperature without reference to any complex terminologies.

Corollary 0.2.1.1 (The First Definition of Temperature)**.** *Every object has a property called* ***temperature****. When two objects are in* ***thermal equilibrium****, their temperatures are* ***equal****.*

After reading the temperature of hot water in our cup (might be around 60°C), you wait for a while, and the mercury level inside thermometer slowly decreases. You carefully observe until the water has completely cooled down to room temperature. Finally, the mercury level stops decreasing, and at this point the room temperature, the water temperature, and the thermometer temperature are all the same (possibly 25°C), according to the law.

0.3 Thermal Expansion and Heat Absorption

Let's look closer to the experiment that we did. If I use a clock to measure time, I will obtain a simple graph:

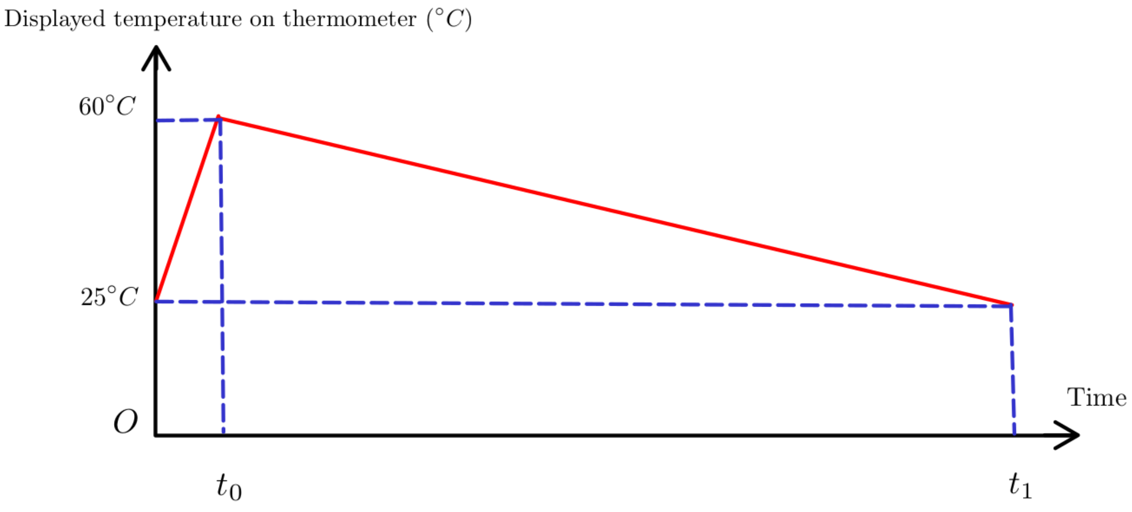


Figure 1: Relationship between time and thermometer's temperature

As you can see here, the time interval $(0, t_0)$ is very short compared to (t_0, t_1). Look at the first interval [1], why did the thermometer's temperature increase?

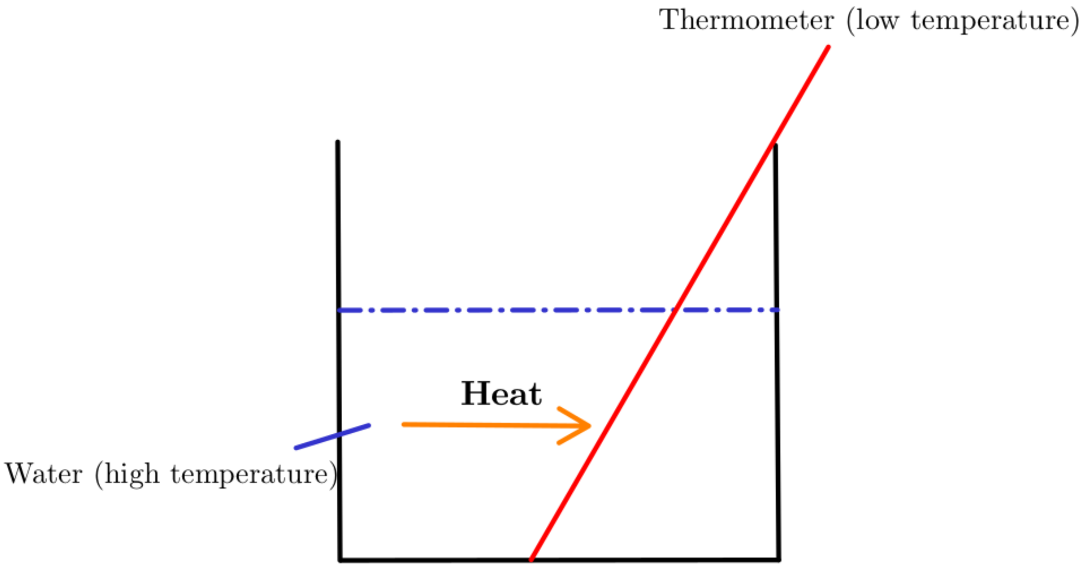


Figure 2: Heat flows between objects

The simplest explanation is that there is a "special energy" that transfers from a high-temperature object to a low-temperature object. This is very easy to imagine, and we call that "energy" as **heat** [2]. Now you might wonder "Why does not heat flow in the reverse direction?". This question will be explained carefully in the next chapters.

Definition 0.3.1 (Heat Definition)**.** *Heat is the energy that **spontaneously** transfers from the higher temperature object to the lower temperature object.*

[1] For a very short period, the amount of energy transferred to the outside environment is negligible, so we can ignore it.

[2] **Heat** in this context is a **positive quantity**. I will explain the negative case later.

Because heat is a **transferring energy** between objects, so we have to be careful here:

Corollary 0.3.0.1. *A thermodynamics system **does not** contain heat.*

Now we can form the second definition of **temperature** bases on Definition 0.3.1 as follows:

Corollary 0.3.0.2 (The Second Definition of Temperature)**.** ***Temperature** is a measure of the tendency of an object to **spontaneously** give up energy to its surroundings.*

Obviously, from Definition 0.3.1 and Corollary 0.3.0.2, if we define Q as heat and T as temperature, we obtain the proportional relationship:

$$Q \propto \Delta T$$

Definition 0.3.2 (Heat Capacity Definition)**.** *The **heat capacity** C [1] of an object is defined as:*

$$Q = C\Delta T = C(T_f - T_i)$$

As you can see here, the unit of quantity C is (J/K)[2]. In practice, it is more convenient to define a new quantity as "heat capacity per mass" c (J/K.kg), and standardize their values into a useful look-up table for searching.

Corollary 0.3.0.3 (Specific Heat Definition)**.** *The specific heat is defined as:*

$$c = \frac{C}{m}$$

Or equivalent to:

$$Q = mc\Delta T$$

Since the initial temperature of hot water is higher than thermometer, hot water tends to give up energy in **heat form** to the thermometer. We can easily calculate how much released energy using the above formula.

Example 0.3.1. How much heat flows from hot water to thermometer? Known that $m_{\text{Hg}} = 0.5$ (g) and $c_{\text{Hg}} = 138$ (J/K.kg).

We have:

$$Q = mc\Delta T^3 = \frac{0.5}{1000}.138.(60 - 25) = 2.415 \text{ (J)}$$

Example 0.3.2. How much heat flows from the thermometer to hot water? What is the temperature of hot water at the moment t_0? Known that $m_{\text{H}_2\text{O}} = 50$ (g) and $c_{\text{H}_2\text{O}} = 4181$ (J/K.kg).

We have to very be careful here. Hot water gives **positive** energy in heat form to thermometer, so it receives **negative** energy flows from thermometer. Formally, we easily represent this relationship:

$$Q_{\text{hot water to thermometer}} = -Q_{\text{thermometer to hot water}}$$

The direction of heat flowing is very significant, that directly rules the **sign** of Q. Eventually, we obtain:

$$Q_{\text{thermometer to hot water}} = -2.415 \text{ (J)}$$

[1] Actually, I think the word "capacity" here is used due to historical reason, and it is not so accurate in physics context.

[2] Up to now, we still not define what K is; but you can think K (Kelvin) as a new temperature scale, like Celsius. The relationship between them is linear: $T_K = T_C + 273°\text{C}$.

[3] Converting Celsius to Kelvin scale does not affect ΔT.

Applying Corollary 0.3.0.3, we see that:

$$Q = mc\Delta T = \frac{50}{1000}.4181.(T_f - 60) = -2.415 \Rightarrow T_f = 59.98^\circ \text{ C}$$

Because ΔT is very low, so we can neglect this tiny changing and shifting our focus on the temperature changing of thermometer.
During the heat absorbing process, the mercury level slowly increases inside our thermometer, but why? Our common belief is that "If something is hot, it will get bigger, and vice versa.". Representing our claim into formulas, we obtain:

$$\Delta L \propto \Delta T$$

$$\Delta V \propto \Delta T$$

with L stands for length, and V stands for volume. In this chapter, we will temporarily accept our assertions (or common beliefs) as true without further explanation. They will be rediscovered in more detail in later chapters.

Definition 0.3.3 (Definition of Linear Expansion)**.** *If the temperature of a metal rod of length L is raised by an amount of ΔT:*

$$\Delta L = L\alpha\Delta T$$

*The constant α is called the **coefficient of linear expansion**.*

Using the same line of thinking, we can extend our Definition 0.3.3 above to volume expansion as a corollary.

Corollary 0.3.0.4 (Definition of Volume Expansion)**.** *If the temperature of solids, or liquids of volume V is raised by an amount of ΔT:*

$$\Delta V = V\beta\Delta T$$

*The constant β is called the **coefficient of volume expansion**.*

Obviously, the relationship between α and β can simply be described as:

Corollary 0.3.0.5.

$$\beta = 3\alpha$$

Both α and β share the same unit 1/K or K^{-1}, and like the specific heat quantity c, these coefficients are found by experiment.

Example 0.3.3. During the first time interval, the mercury temperature slowly increase from $T_i = 25°C$ to $T_f = 60°C$. Known that $\beta = 1.8.10^{-4}$ (K^{-1}), and the initial height of mercury level at T_i is $H_i = 10$ (cm), what is the height of mercury level at the moment t_0?

Notice that $V = \pi r^2 H \Rightarrow V \propto H$. Directly using the formula, we see that:

$$\Delta V = V\beta\Delta T \Rightarrow \Delta H = H\beta(T_f - T_i) = \frac{10}{100}.1.8.10^{-4}(60 - 25) = 6.3.10^{-4} \text{ (m)} = 0.063 \text{ (cm)}$$

The final height of mercury level is:

$$H_f = H_i + \Delta H = 10 + 0.063 = 10.063 \text{ (cm)}$$

To conclude this chapter, I would like to focus on a very special case, in which the heat capacity $C = +\infty$. This normally happens at a **phase transformation**, such as melting ice or boiling water.

$$C = \frac{Q}{\Delta T} = \frac{Q}{0} = +\infty$$

We still might want to know how much heat is required to melt or boil the substance completely during a phase transformation. This amount of energy is called the **latent heat**, denoted as L. Similar to the definition of specific heat, we define the quantity **specific latent heat**, denoted by l:

$$l = \frac{L}{m} = \frac{Q}{m} \text{ (J/kg)}$$

Example 0.3.4. How much ice should be added to a cup of 200 grams of boiling water to bring it down to 65°C. Assume that the ice is initially at −15°C, specific heat capacity of ice is 2.1 J/K.g, of water is 4.184 J/K.g; specific latent heat for melting ice is 333 J/g[1].

The flow of heat between ice and boiling water can be represented by formula:

$$Q_{\text{water}\to\text{ice}} = -Q_{\text{ice}\to\text{water}}$$

Or by the **law of conservation of energy** form:

$$Q_{\text{water}\to\text{ice}} + Q_{\text{ice}\to\text{water}} = 0 \text{ (internal energy is conserved, no external energy exists)}$$

Because 65°C is much higher than the freeze point of ice; when the system is equilibrium, all ice is **melted**.

$$\begin{aligned}
Q_{\text{water}\to\text{ice}} &= -Q_{\text{ice}\to\text{water}} \\
m_{\text{H}_2\text{O}} c_{\text{H}_2\text{O}} \Delta T_{\text{H}_2\text{O}} &= -(m_{\text{ice}} c_{\text{ice}} \Delta T_{\text{ice}} + m_{\text{ice}} l + m_{\text{ice}} c_{\text{H}_2\text{O}} \Delta T_{\text{melted ice raising temperature}}) \\
200.4.184.(65-100) &= -[m_{\text{ice}}.2.1.(0-(-15)) + m_{\text{ice}}.333 + m_{\text{ice}}.4.184.(65-0)] \\
\Rightarrow m_{\text{ice}} &= 46.01 \text{ (g)}
\end{aligned}$$

[1] In practice, instead of using standard SI units, we often use their variances like J/g, cal/g,... Chemists really love using gram.

Chapter 1

The First Law of Thermodynamics

1.1 The Microscopic Model

As you can see here, we are moving further than the zeroth law of thermodynamics. In the previous chapter, we neither know what is inside the substances, nor the essence of heat. Thinking like a physicist, we need to build a physical model to explain many phenomena in nature.

Substance is just the collection of many very tiny "imaginary" particles.

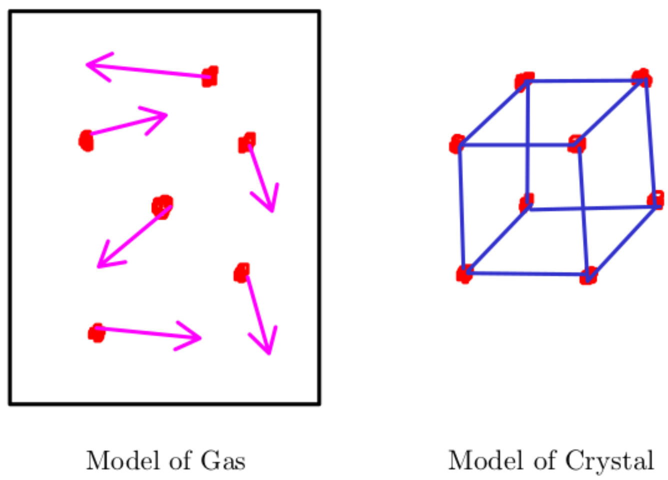


Figure 1.1: Our hypothesis model of gas and crystal

But why do I call these particles "imaginary"? Because in the 19th century, scientists had no way of directly observing them. Physicists living that era built the entire foundation of thermodynamics on atomic theory, which may not have convinced everyone until the quantum age. In the early 20th century, the existence of atoms was finally proven and widely accepted, supported by numerous undeniable physical and mathematical proofs. Finally, in 1955, the first photograph of atom was taken by Erwin W.Muller and his student.

Theorem 1.1.1 (Atomic Theory)**.** *Atomic theory is the specific theory that matter is composed of particles called* ***atoms****.*

I think Theorem 1.1.1 should be called axiom, in mathematical language. In the following chapters, we will focus in depth on how this theory can be applied to build the foundations of thermodynamics, rather than proving it correctness.

1.2 Volume and Pressure

I think everyone knows what volume is, but it is not so easy to formalize our sense to a definition. Mathematically, we can state that:

Definition 1.2.1 (The First Definition of Volume)**.** *Volume is a measure of regions in three-dimensional space.*

But I think we can extend the definition above with atomic theory:

Definition 1.2.2 (The Second Definition of Volume)**.** *Volume is a measure of the amount of space occupied by many small particles of* ***the same type*** *at equilibrium state.*

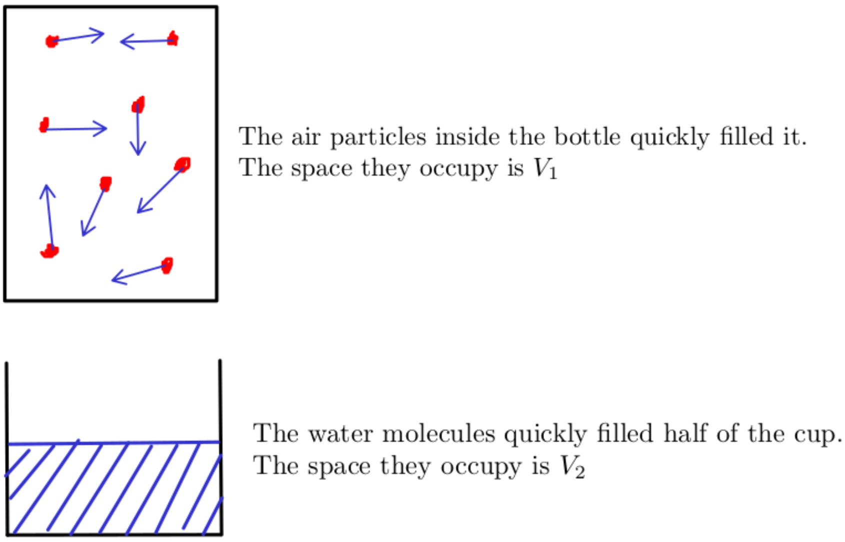


Figure 1.2: Definition of Volume

Now you might wonder if you mix 2 types of completely different gases, such as H_2 and He, are they the same type? The answer is yes, but until they reach the **equilibrium state**.

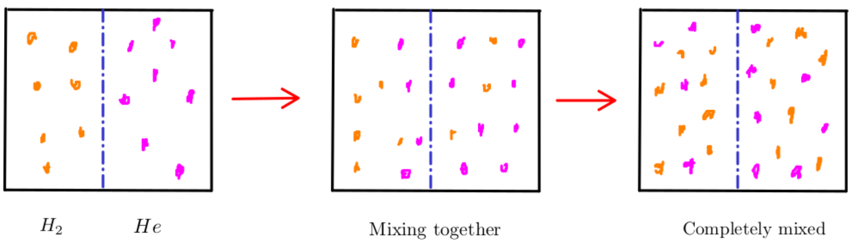


Figure 1.3: Mixing H_2 and He together

Reaching the equilibrium state means the density of gas mixture is the same at all locations inside the box.[1][2]

[1] This is just my claim.

[2] Mixing gas process can release an amount of heat energy. According to the zeroth law of thermodynamics, equilibrium state is reached when the temperature of the mixture is stable.

Unlike volume, **pressure** is very easy concept to define:

Definition 1.2.3 (The First Definition of Pressure)**.** *Pressure is the force applied* ***perpendicular*** *to the surface of an object per unit area over which that force is distributed:*

$$P = \frac{F}{A} \; (N/m^2), \text{ or } (Pa)$$

One classic example of pressure is pressing your hand against a small piston filled with gas. If you press hard and fast enough, you can observe small sparks inside the piston. This phenomenon can be explained by the first law of thermodynamics.

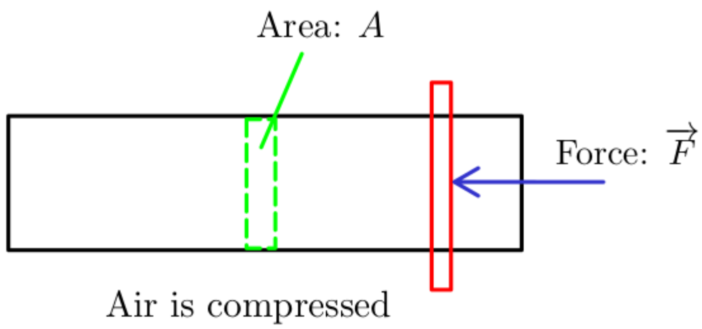


Figure 1.4: Pressing a small piston

After understanding what volume and pressure are, now it is time to answer our previous question from Chapter 0, what is Kelvin scale? Imagine you have an elastic balloon, which can be easily stretched. Of course, this balloon is filled fully with gas.

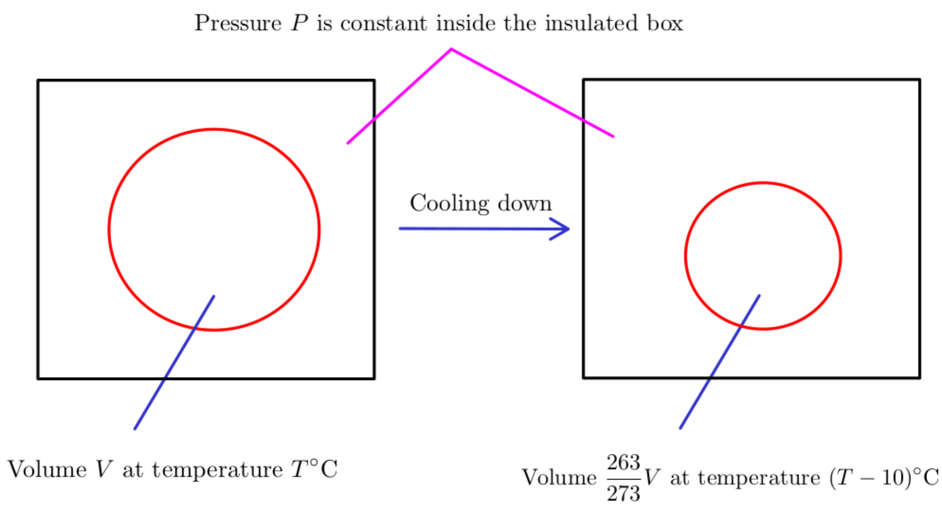


Figure 1.5: For every degree decrease, the volume $V/273$ decreases correspondingly

But because the volume quantity can not go below zero, if the temperature reaches $(T - 273)$°C, and continues to decrease, the volume will remain at zero. Alternatively, if you hold the volume of the gas fixed, then its pressure will approach zero as the temperature approaches $(T-273)$°C. The special value -273°C is called **absolute zero**[1], and defines the initial point of the **absolute temperature scale**, named after Lord Kelvin[2].

Definition 1.2.4 (Definition of Kelvin Scale)**.** *The Kelvin scale is an absolute temperature scale that starts at the lowest possible temperature, taken to be* $0K$*:*

$$T_K = T_C + 273°C \ (K)$$

1.3 The Ideal Gas Law

Discovered by experiments, many properties of a low-density gas can be summarized in the famous idea gas law, where P (N/m^2), V (m^3), n (mole), R is a universal constant, and T (K).

Theorem 1.3.1 (The Ideal Gas Law)**.** *The empirical ideal gas law for* ***low-density gas*** *is:*

$$PV = nRT$$

This section does not have much to discuss beyond introducing an important law of gases and some new constants. As you can see here, the unit of constant R is:

$$R = 8.31 \text{ (J/mol.K)}$$

If we define $N_A = 6.023.10^{23}$ (mol^{-1}) is Avogadro's number, then the law can be rewritten as:

$$PV = nRT = \frac{N}{N_A}RT = N\mathbf{k_B}T$$

The constant k_B is called **Boltzmann's constant**, it plays vital role in thermodynamics:

$$k_B = \frac{R}{N_A} = 1.381.10^{-23} \text{ (J/K)}$$

1.4 The Kinetic Theory of Gases

In a gas-filled piston, you already know that gas (or any substance) is simply a collection of tiny, chaotic, uncontrollable particles. If you heat the piston, these particles appear to move faster. In this section, we will use the gas model to quantify the relationship between piston temperature and the velocity of the particles.

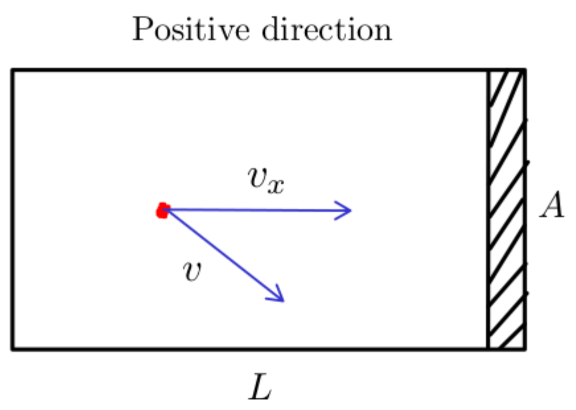


Figure 1.6: An oversimplified model of an ideal gas

[1]This is the basic idea of how absolute zero was discovered.

[2]How about holding both the volume and pressure of the gas fixed, and decrease temperature?

Since the system of 10^{23} particles is impossible to process in the first place, I have to make some assumptions here to simplify our physical model:

- My oversimplified model of an ideal gas contains only 1 particle.
- Its speed never changes.
- Piston is perfectly smooth, so collisions are always **elastic**.

Directly applying Definition 1.2.3 yields:

$$\overline{P}_{\text{x, on piston}} = \frac{\overline{F}_{\text{x, on piston}}}{A} = -\frac{\overline{F}_{\text{x, on particle}}}{A} \text{ (Newton's Third Law)}$$
$$= -\frac{m\dfrac{\Delta v}{\Delta t}}{A} = -\frac{m\dfrac{(-v_x)-v_x}{2L/v_x}}{A} = \frac{mv_x^2}{V}$$

If our piston contains N particles, then the total pressure on piston by x-axis is:

$$P_x = \sum \overline{P}_{\text{x, on piston}} = \frac{\sum mv_{x,i}^2}{V} = N\frac{m\overline{v_x^2}}{V} \Rightarrow P_xV = Nm\overline{v_x^2}$$

Because N is extremely large, and the gas is homogeneous:

$$P = P_x = P_y = P_z \Rightarrow PV = Nm\overline{v_x^2}$$

The average velocity of the particles in the x, y and z directions is the same, we obtain:

$$\frac{m\overline{v_x^2}}{2} = \frac{m\overline{v_y^2}}{2} = \frac{m\overline{v_z^2}}{2} = \frac{k_BT}{2}$$

The average velocity of N particles in three-dimensional space is defined as:

$$\overline{v^2} = \overline{v_x^2 + v_y^2 + v_z^2} = \frac{v_x^2 + v_y^2 + v_z^2}{N} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}$$

The average translational kinetic energy is then:

$$\overline{K} = \frac{1}{2}m\overline{v^2} = \frac{1}{2}m\overline{v_x^2} + \frac{1}{2}m\overline{v_y^2} + \frac{1}{2}m\overline{v_z^2} = \frac{3k_BT}{2}$$

We can simply explain the phenomenon of thermal expansion using the previous result. If the temperature increases, the average kinetic energy also increases, which means these particles tend to move faster. Their chemical bonds tend to loosen or break, which causes the length or volume of matter to increase slightly.

$$\overline{v^2} = \frac{3k_BT}{m}$$

We are also interested in the **average of the squares of the speeds** (or root-mean square, rms for short):

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_BT}{m}}$$

As you can see here, we have just done a bit of Statistical Physics. Because we do not know the **exact** state of each particle in a set of 10^{23} particles, we can still calculate their average velocity or energy[1]. In the spirit of Statistical Physics, we solve the whole big problem of the enormous number of observations, without knowing the behavior or each individual particle.

[1]Potential energy $U = mgz$ can be ignored, since $K >> U$.

1.5 The Equipartition Theorem

The average translational kinetic energy is just a special case of a much more general result. In the previous discussion, we only considered the motion of particles moving in three-dimensional space. There are 3 types of motion, directly related to 3 types of kinetic energy, and each of them can be called a **degree of freedom**:

1. Moving along x-axis: $\overline{K}_x = \frac{1}{2}m\overline{v_x^2} = \frac{k_B T}{2}$
2. Moving along y-axis: $\overline{K}_y = \frac{1}{2}m\overline{v_y^2} = \frac{k_B T}{2}$
3. Moving along z-axis: $\overline{K}_z = \frac{1}{2}m\overline{v_z^2} = \frac{k_B T}{2}$

Other degrees of freedom include rotational motion and elastic potential energy (as stored in a structure like a spring), and what they have in common is that their energy functions are all **quadratic form**.

Theorem 1.5.1 (The Equipartition Theorem)**.** *At temperature T, the average energy of any quadratic degree of freedom is $\frac{1}{2}k_B T$.*

Corollary 1.5.1.1 (Total Internal Energy)**.** *If a system contains N particles, each with f degrees of freedom, and the **total internal energy**[1] is:*

$$U = N.f.\frac{1}{2}k_B T$$

The final problem is determining the value of f. For example, if our gas is monatomic, then we can easily conclude that $f = 3$. In the case of diatomic, $f = 5$ because there are 5 possible ways of motion:

1. Moving along x-axis: $\overline{K}_x = \frac{1}{2}m\overline{v_x^2} = \frac{k_B T}{2}$
2. Moving along y-axis: $\overline{K}_y = \frac{1}{2}m\overline{v_y^2} = \frac{k_B T}{2}$
3. Moving along z-axis: $\overline{K}_z = \frac{1}{2}m\overline{v_z^2} = \frac{k_B T}{2}$
4. Rotating around x-axis: $\overline{K}_{x,\text{ rotate}} = \frac{1}{2}I\omega_x^2 = \frac{k_B T}{2}$
5. Rotating around y-axis: $\overline{K}_{y,\text{ rotate}} = \frac{1}{2}I\omega_y^2 = \frac{k_B T}{2}$

Most polyatomic molecules can rotate about all three axes, so if our gas is polyatomic, then $f = 6$. In the case of solid, $f = 6$ too.

[1] Actually, there is a slight difference between total thermal energy and total internal energy; but I neglect them and keep it simple as possible.

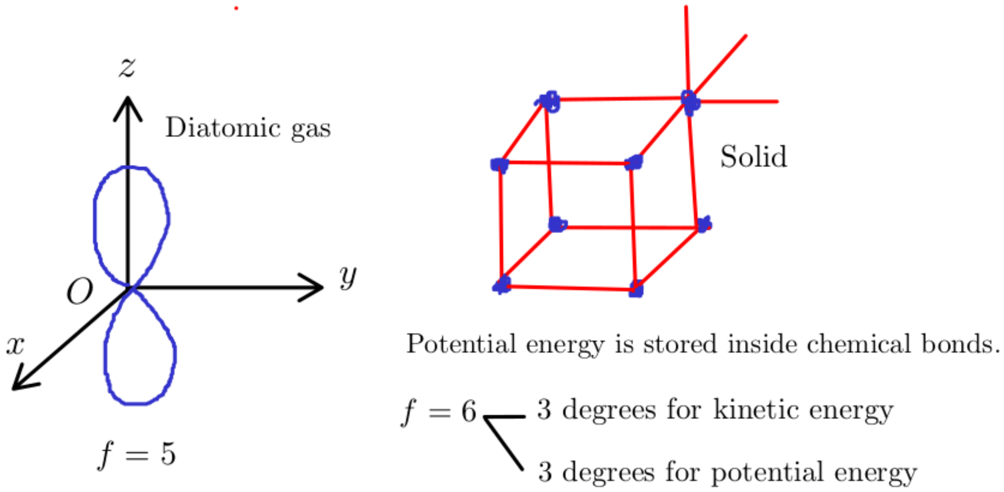


Figure 1.7: Degrees of freedom illustration

Counting degrees of freedom of liquid is more complicated than gas or solid, since the intermolecular potential energies are not quadratic functions.

Example 1.5.1. Calculate the internal energy of a bottle filled with H_2 at normal room temperature.

Assuming the normal room condition are $P = 1$ (atm), $T_C = 25°\text{C}$ and the bottle's volume is $V = 500$ (ml). Using the idea gas law, we have:

$$PV = Nk_BT \Rightarrow N = \frac{PV}{k_BT}$$

Applying Corollary 1.5.1.1, we obtain:

$$U = N.f.\frac{k_BT}{2} = \frac{PV}{k_BT}.f.\frac{k_BT}{2} = \frac{PVf}{2} = \frac{101325.0.5.10^{-3}.5}{2} = 126.65 \text{ (J)}$$

1.6 The First Law of Thermodynamics

1.6.1 Heat and Work

Consider a piston filled with ideal gas[1], now you heat it up with a candle, what will happen?

[1] Actually, you do not need to worry too much about the word "ideal", because most of the gases we encounter in real life behave similar to ideal gas model.

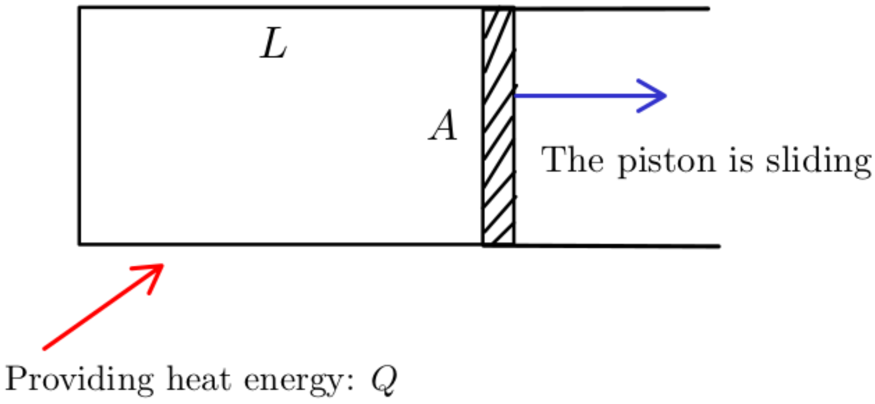


Figure 1.8: Heat Q is converting to work W to push the piston away

I think we are all familiar with the definition of work in Classical Mechanics course:

$$W = \int_{\overrightarrow{r_i}}^{\overrightarrow{r_f}} \overrightarrow{F} d\overrightarrow{r} \text{ (J)}$$

Both work W and heat Q are 2 types **energy**; but they have a slight but subtle difference:

- Heat is defined as any **spontaneously flow of energy** from one object to another. As we discussed in the previous chapter, heat tends to flow from the higher temperature object to the lower, but **not the reverse direction**. We have not known the reason yet.

- Work is defined as any **other transfer of energy into or out of a system** (not spontaneous).

Notice that both heat and work refer to energy in **transmit**. It is meaningless to ask how much heat or work inside the system. We can only talk about how much heat or work have entered or left the system. According to the law of conservation of energy, energy can only be transformed or transferred from one to another. If we use the symbols Q and W to represent the amounts of energy that **enter** a system as heat and work; U as the internal energy of system, then we can confidently state that:

External energy will be transformed into an increase in internal energy.

Or written as an equation, this statement is:

$$\Delta U = Q + W^1$$

Theorem 1.6.1 (The First Law of Thermodynamics)**.** *The increment in the internal energy is* ***equal*** *to the difference between heat accumulated by the system and the thermodynamics work done by it.*

$$\Delta U = Q - W^2$$

In this scope of this book, we will stick to the convention that I have mentioned above, to keep consistent, that means W is representing the amount of energy that **enter** the system.

[1]Many other textbooks, including Halliday, define W as work that **left** a system, so the sign of W is negative:

$$\Delta U = Q - W$$

This convention works very great when solving heat engine or heat cycle problems.

[2]I read this statement from Wikipedia, and I think this is the best explanation for the first law.

1.6.2 Thermodynamics Process

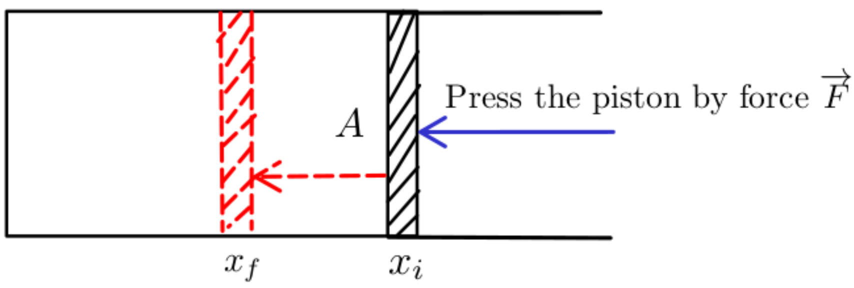


Figure 1.9: Piston is being compressed by force

Using the definition of work, we obtain:

$$W = \int_{\overrightarrow{r_i}}^{\overrightarrow{r_f}} \overrightarrow{F} d\overrightarrow{r} = \int_{x_i}^{x_f} \overrightarrow{F} d\overrightarrow{x} = \int_{x_i}^{x_f} F \cos(\overrightarrow{F}, d\overrightarrow{x}) dx \text{ (Moving along } x\text{-axis)}$$

If we define W_{left} as the work that left our piston, then:

$$W_{\text{entered}} = -W_{\text{left}} = -\int_{x_i}^{x_f} F \cos(\overrightarrow{F}, d\overrightarrow{x}) dx = -\int_{x_i}^{x_f} F dx = -\int_{x_i}^{x_f} PA dx = -\int_{V_i}^{V_f} PdV$$

For convenience, W denotes the work that **entered** system:

$$W = -\int_{V_i}^{V_f} PdV$$

Within the scope of this subsection, I will list 4 main special thermodynamics processes, without going deeply in details.

Isotherm Process

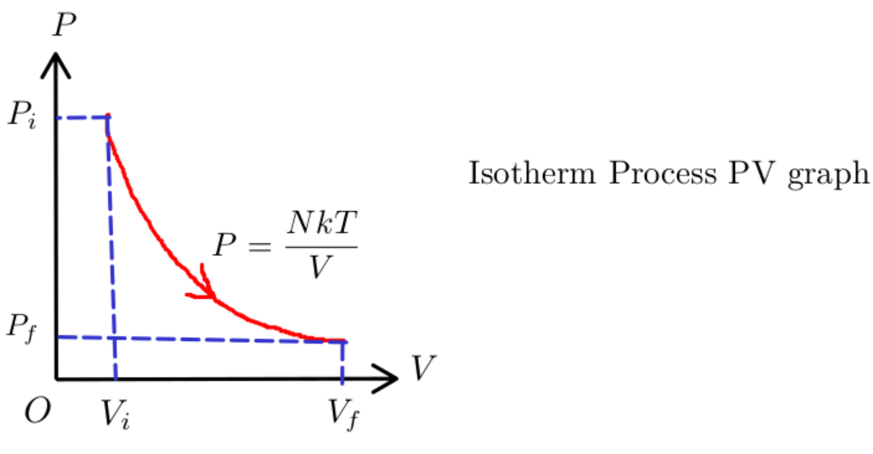


Figure 1.10: Isotherm PV graph

From Corollary 1.5.1.1, we can see that:

$$\Delta U = N.f.\frac{1}{2}k_B\Delta T$$

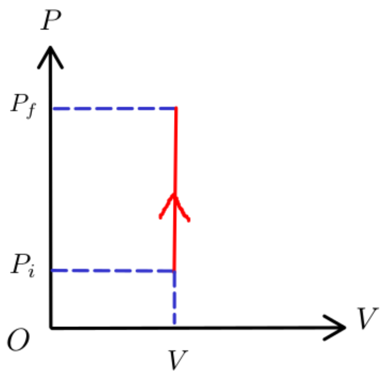
Because T is constant, we can conclude that:

$$\Delta U = Q + W = 0 \Rightarrow Q = -W$$

The work done on the system is:

$$W = -\int_{V_i}^{V_f} PdV = -\int_{V_i}^{V_f} \frac{Nk_BT}{V}dV = Nk_BT\ln\frac{V_i}{V_f}$$

Constant-Volume Process



Constant-Volume Process PV graph

Figure 1.11: Constant-Volume PV graph

According to the first law:

$$\Delta U = Q + W = Q - \int_{V_i}^{V_f} PdV = Q \text{ (no work is done)}$$

Recall the definition of heat capacity from Definition 0.3.2:

$$C = \frac{Q}{\Delta T} = \frac{\Delta U - W}{\Delta T}$$

We define a new quantity, is called **heat capacity at constant volume**, denoted C_V[1,2]:

$$C_V = \frac{\Delta U - W}{\Delta T} = \frac{\Delta U}{\Delta T} = \left(\frac{\partial U}{\partial T}\right)_V = \frac{Nk_Bf}{2}$$

C_V is so convenient when we want to calculate the amount of internal energy increasing (ΔU), or heat Q entering the system:

$$\Delta U = Q = C_V\Delta T = \frac{Nk_Bf\Delta T}{2}$$

[1]Many other textbooks, including Halliday, define the quantity C_V as **molar specific heat at constant volume:**

$$C_V = \frac{Q}{n\Delta T}$$

There is no conflict here, this convention works very well in solving physics exercises; however the capital C might be confused with the definition of thermal capacity, as we discussed above.

[2]I am following the notation convention from "An Introduction to Thermal Physics", written by Daniel V. Schroeder.

Constant-Pressure Process

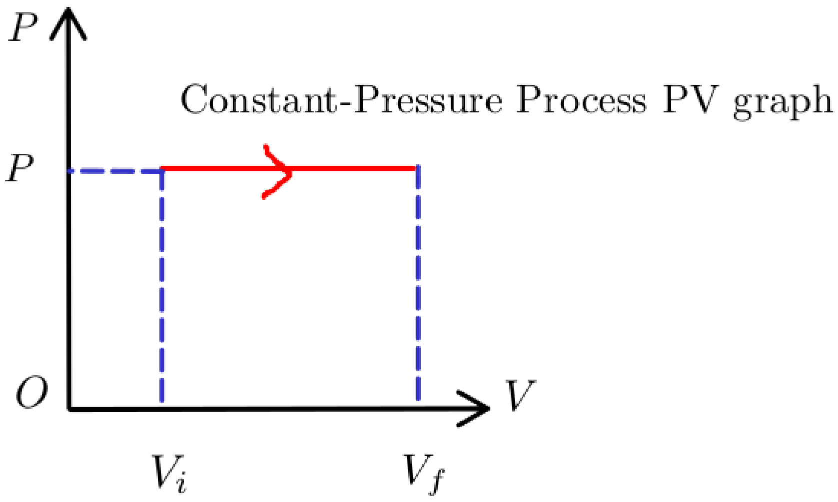


Figure 1.12: Constant-Pressure PV graph

Like C_V, we denote C_P as **heat capacity at constant pressure**, which can be determined as:

$$\begin{aligned} C_P &= \frac{Q}{\Delta T} = \frac{\Delta U - W}{\Delta T} = \frac{\Delta U + P\Delta V}{\Delta T} = C_V + P\left(\frac{\Delta V}{\Delta T}\right) \\ &= C_V + P\left(\frac{\partial V}{\partial T}\right)_P = C_V + Nk_B \end{aligned}$$

The increase of internal energy is:

$$\Delta U = C_V \Delta T = \frac{Nk_B f \Delta T}{2}$$

The work done and heat that entered the system are:

$$\begin{aligned} W &= -P\Delta V \\ Q &= C_P \Delta T = (C_V + Nk_B)\Delta T \end{aligned}$$

Adiabatic Process

The adiabatic compression is the process that **no heat flows out of (or into) the gas**. I always think of this process within the Carnot cycle[1] as something happens very quickly and no heat is leaked or entered into the system.

[1]Carnot cycle in particular, and heat engines in general, will be analyzed in more detail in later chapters.

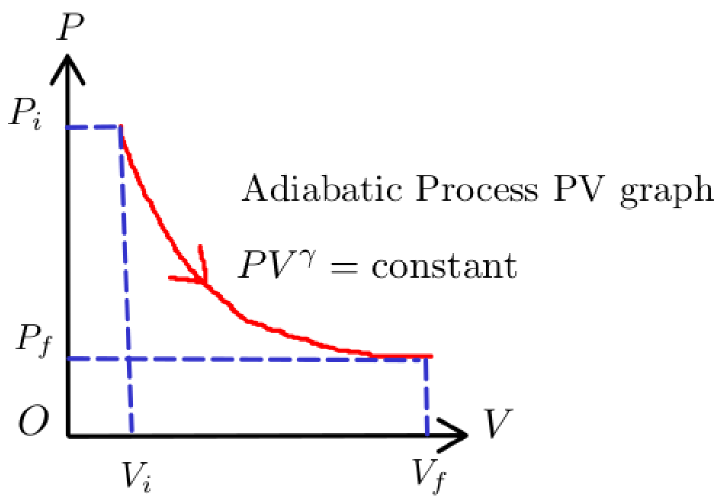


Figure 1.13: Adiabatic PV graph

Because heat is not leaked or entered into system:

$$\Delta U = Q + W = 0 + W = W$$

The original form of this equation does not reveal any information about the relationship between P and V themselves, independent of T:

$$\Delta U = W = -\int_{V_i}^{V_f} PdV \Rightarrow dU = -PdV \text{ (Differentiate both sides)}$$

Using Corollary 1.5.1.1, we can see that:

$$\begin{aligned} dU &= -PdV \\ \Leftrightarrow \frac{Nk_B f}{2}.dT &= -PdV \\ \Leftrightarrow \frac{Nk_B f}{2}.dT &= -\frac{Nk_B T}{V} dV \\ \Leftrightarrow \int \frac{Nk_B f}{2}.\frac{dT}{T} &= -\int Nk_B.\frac{dV}{V} \text{ (Take antiderivative both sides)} \\ \Leftrightarrow \frac{f}{2} \ln T &= -\ln V + C \\ \Leftrightarrow T^{f/2} V &= \text{constant} \end{aligned}$$

Here is a very tricky step[1][2], power both sides by $2/f$:

$$(T^{f/2}V)^{2/f} = TV^{2/f} = \frac{PV}{Nk} V^{2/f} = \text{constant} \Rightarrow PV^{(f+2)/f} = \text{constant}$$

If we define the coefficient γ as **adiabatic exponent**:

$$\gamma = \frac{f+2}{f} = \frac{C_P}{C_V}$$

then we can rewrite the $P - V$ adiabatic function:

$$PV^\gamma = \text{constant}$$

[1]If you find this proof very unnature, so do I. I recalculated the power factor $2/f$ from the solution, and it is very difficult to find on the first approach to the problem.

[2]If you are still not satisfied, I recommend that you should read the proof in section 19-9, Halliday textbook.

Chapter 2

The Second Law of Thermodynamics

Ludwig Boltzmann, who spent much of his life studying statistical mechanics, died in 1906, by his own hand. Paul Ehrenfest, carrying on the work, died similarly in 1933. Now it is our turn to study statistical mechanics[1].

2.1 Microstate and Macrostate

In the first chapter, we used an ideal statistical model to develop the kinetic theory of gases and obtained some very useful results that formed the basis of the first law of thermodynamics. Now, we continue to use our knowledge of probability and statistics to answer our question: "Why does heat[2] flow from high to low temperatures, but not in the reverse direction?".

Let's do some combinatorics here. Imagine you have a fair coin, and you flip that coin 3 times. You define an event E as "Landing 2 heads". How many sample points are there inside the subset E?

$$E = \{THH, HTH, THH\} \Rightarrow n(E) = \binom{3}{2} = 3$$

How many possible outcomes are there inside the sample space Ω (or S)?

$$\Omega = \{TTT, TTH, THT, THH, HHH, HHT, HTH, HHH\} \Rightarrow n(\Omega) = 2^3 = 8$$

The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the macrostate "Landing 2 heads" has 3 microstates, one of them is THH; the number of microstates corresponding to a macrostate is called the **multiplicity**. Now I will change the notation here, I use the symbol Ω to denote **multiplicity**[3]. Generally, the multiplicity of the macrostate "Landing n heads" when flipping a coin N times is:

$$\Omega(N, n) = \binom{N}{n}$$

2.2 The Einstein Model of a Solid

Actually, the Einstein model is quite difficult for me to fully understand it in a quantum context; so I just imagine it as a solid where every chemical bond (mostly ionic bond) stores some amount of energy, that shares the same unit. Let me explain.

[1] Very famous quote that I found on the internet.

[2] Heat in this context, is a **positive quantity.**

[3] You can relate this new convention to the **random variable** concept in math books.

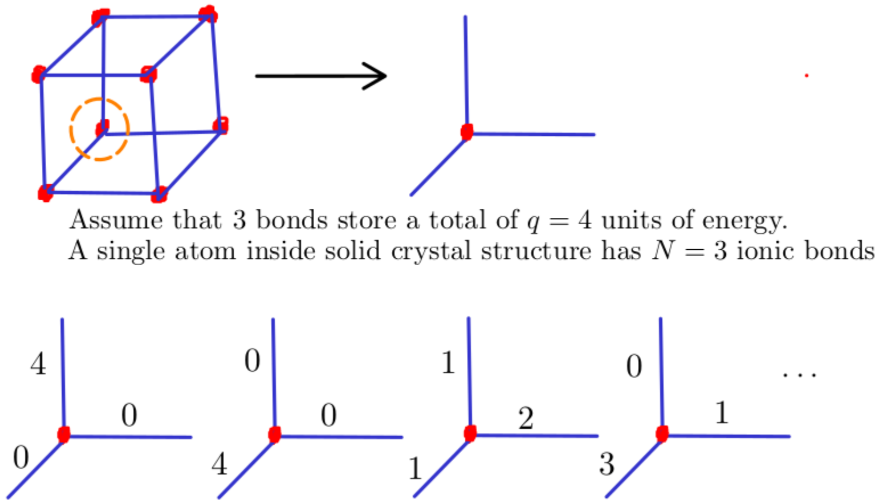


Figure 2.1: How many configurations satisfy 3 bonds containing 4 units of energy?

A solid contains N chemical bonds[1,2], and store a total of q units of energy. The multiplicity is simply the number of **non-negative** solutions of this equation:

$$x_1 + x_2 + \cdots + x_{N-1} + x_N = q$$

Or equivalently, the number of each solution must be **greater than or equal to** 1 of this equation:

$$(x_1 + 1) + (x_2 + 1) + \cdots + (x_{N-1} + 1) + (x_N + 1) = q + N$$

$$\Leftrightarrow t_1 + t_2 + \cdots + t_{N-1} + t_N = q + N$$

This combinatorial problem is very classic; we just need to "split" $(q+N)$ units of energy using $(N-1)$ "bars".

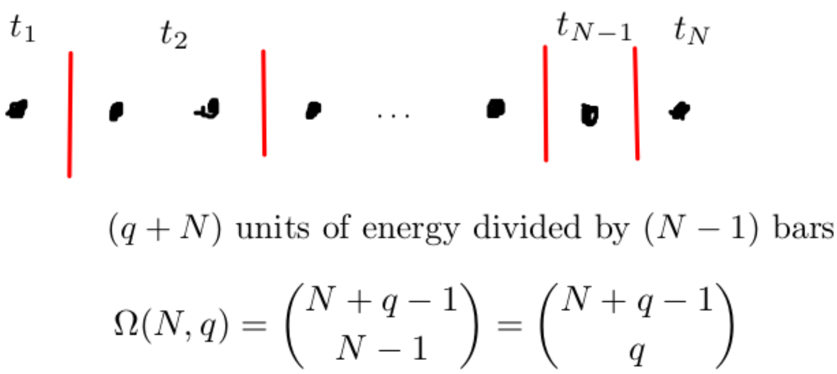


Figure 2.2: The general formula for the multiplicity of an Einstein solid

[1] I think it is more accurate to use term "chemical bonds", rather than "oscillators" here.

[2] As of the time of writing, I still do not understand why solids have 6 degrees of freedom. Atoms in a crystal lattice barely move, so logically, their kinetic energy should be zero, and $f = 3$ is simply due to potential energy stored inside the ionic bonds?

This crucial formula is the backbone of this section, especially when we deal with problems related to Einstein's solid model.

Theorem 2.2.1 (The Multiplicity of an Einstein solid)**.** *The general formula for the multiplicity of an Einstein solid with N chemical bonds and q units of energy is:*

$$\Omega(N,q) = \binom{N+q-1}{q} = \frac{(N+q-1)!}{(N-1)!q!}$$

In our previous example in Figure 2.1, the number of satisfied configurations is:

$$\Omega(N,q) = \Omega(3,4) = \binom{3+4-1}{4} = 15$$

The problem is, N and q are always very large numbers. For example, in 1 gram of Na, the number of chemical bonds is up to $N = 7.856.10^{22}$. In statistical physics, we usually deal with very large number; therefore, in the remainder of this subsection, I will introduce some powerful mathematical tools to handle them easier.

Theorem 2.2.2 (Weak Stirling's Approximation for Natural Logarithm)**.** *When $N >> 1$:*

$$\ln N! = N \ln N - N$$

Proof.

$$\begin{aligned}\ln N! &= \ln\left[N(N-1)\cdots 2.1\right] = \ln N + \ln(N-1) + \cdots + \ln 2 + \ln 1 \\ &= \sum_{i=1}^{N} \ln i \approx \int_1^N \ln x dx \approx N \ln N - N\end{aligned}$$

□

Theorem 2.2.3 (Stirling's Approximation)**.** *When $N >> 1$:*

$$N! \approx N^N e^{-N}\sqrt{2\pi N}$$

Proof. Recall the definition of Gamma function, for every non-negative integer N, we always have:

$$\begin{aligned}\Gamma(N+1) &= \int_0^{+\infty} x^N e^{-x} dx = N! \\ &= \int_0^{+\infty} e^{N\ln x} e^{-x} dx = \int_0^{+\infty} e^{N\ln x - x} dx \\ &= \int_0^{+\infty} e^{N\ln x - x} dx\end{aligned}$$

Now we want to expand the equation by Taylor expansion, by setting an auxiliary variable $x = y + N \Rightarrow y = x - N$:

$$\begin{aligned}N\ln x - x &= N\ln(y+N) - (y+N) \\ &= N\ln N\left(\frac{y}{N}+1\right) - (y+N) \\ &= N\ln N + N\left(\frac{y}{N} - \frac{1}{2}.\frac{y^2}{N^2}\right) - y - N \\ &= N\ln N - \frac{1}{2}.\frac{y^2}{N} - N\end{aligned}$$

After changing the variable, we obtain:

$$\Gamma(N+1) = \int_0^{+\infty} e^{N\ln x - x} dx = \int_{-N}^{+\infty} \exp\left(N\ln N - \frac{1}{2}.\frac{y^2}{N} - N\right) dy$$
$$= N^N e^{-N} \int_{-N}^{+\infty} \exp\left(-\frac{1}{2}.\frac{y^2}{N}\right) dy$$

Since N approaches infinity, applying Gaussian integral result:

$$\int_{-\infty}^{+\infty} e^{-y^2} dy = \sqrt{\pi}$$

We conclude:

$$\Gamma(N+1) = N^N e^{-N}\sqrt{2N} \int_{-\infty}^{+\infty} \exp\left[-\left(\frac{y}{\sqrt{2N}}\right)^2\right] d\left(\frac{y}{\sqrt{2N}}\right)$$
$$= N^N e^{-N}\sqrt{2\pi N}$$

□

Let's apply our previous results to explore a very cool phenomenon. I have an **interacting system**, which consists 2 Einstein solids. I call them A and B; solid A contains 2 atoms and solid B contains 3 atoms. From the simple cube crystal structure, which I drew above, we know that $N_A = 6$ and $N_B = 9$ chemical bonds[1]. You measure the system's **temperature** through the insulation box, and obtain the total **internal energy** result $q = q_A + q_B = 8$ (units of energy)[2]. How is the energy distributed inside the solid A or B? I think picking them out of the box might be not a good idea, since they can immediately interact with the outside environment.

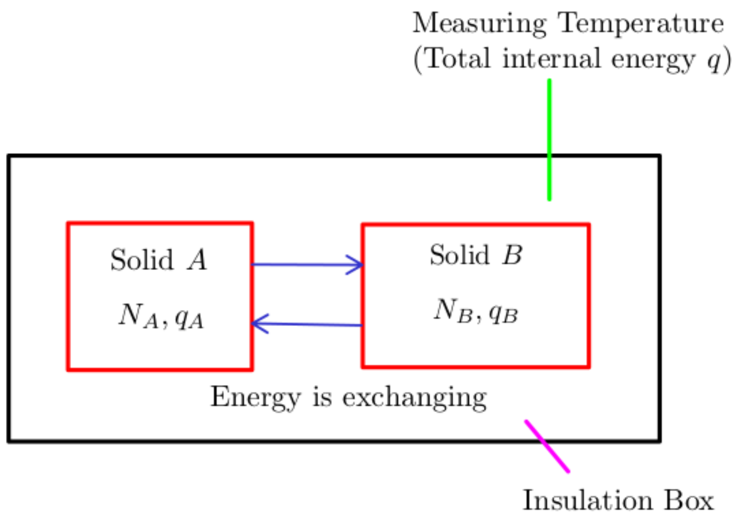


Figure 2.3: What is the most probable configuration for energy distributing system?

[1]In practical, I think we should determine the number of moles first, then search for the actual crystal structure. Cube is just the simplest case.

[2]This is just a very ideal model experiment.

We have to do an indirect way to consider how many units of energy are likely within our solids. Recall Theorem 2.2.1, the number of total possible states of a solid of N chemical bonds with q energy units is:

$$\Omega(N,q) = \binom{q+N-1}{q}$$

We already assumed that $N_A = 6$, $N_B = 9$, and $q = q_A + q_B = 8$, so:

$$\Omega(N_A, q_A) = \Omega(6, q_A) = \binom{q_A+5}{q_A}$$

$$\Omega(N_B, q_B) = \Omega(9, q_B) = \binom{q_B+8}{q_B}$$

The multiplicity of an interacting system is: $\Omega_{AB} = \Omega_A.\Omega_B$

q_A	Ω_A	q_B	Ω_B	Ω_{AB}
0	1	8	12870	12870
1	6	7	6435	38610
2	21	6	3003	63063
3	56	5	1287	72072
4	126	4	495	62370
5	252	3	167	42084
6	462	2	45	20790
7	792	1	9	7128
8	1287	0	1	1287

As you can see here, the most probable configuration of an interacting system is $q_A = 3$, $q_B = 5$, with the highest multiplicity, the corresponding probability is slightly higher than its surroundings:

$$P(q_A = 2, q_B = 6) = 19.69\%$$

$$P(q_A = 3, q_B = 5) = 22.50\%$$

$$P(q_A = 4, q_B = 4) = 19.47\%$$

Our calculation above seems like make sense; the "equilibrium" state of energy is the most likely reached state. Energy is well-distributed inside solids, represented by a nearly perfect ratio:

$$\frac{q_A}{q_B} \approx \frac{N_A}{N_B} \Rightarrow \frac{q_A}{N_A} \approx \frac{q_B}{N_B}$$

The amount of energy stored is directly proportional to the number of chemical bonds in a solid; to put it more simply, the more containers, the more energy can be stored. Now consider another situation where $N_A = N_B = 300$, and $q = q_A + q_B = 1000$. Instead of directly calculating $\Omega(N,q)$, we must change our strategy to **approximate** numerical values.

$$\Omega(N,q) = \binom{q+N-1}{q} \approx \binom{q+N}{q} = \frac{(q+N)!}{q!N!} \quad (q, N >> 1)$$

Take natural logarithm of both sides, and use Theorem 2.2.2:

$$\begin{aligned}\ln\Omega(N,q) &= \ln{(q+N)!} - \ln q! - \ln N! = (q+N)\ln{(q+N)} - (q+N) - q\ln q + q - N\ln N + N \\ &= (q+N)\ln{(q+N)} - q\ln q - N\ln N\end{aligned}$$

Taking natural logarithm of the multiplicity of an interacting system yields:

$$\ln\Omega_{AB} = \ln\Omega_A + \ln\Omega_B$$

Obviously, if Ω_{AB} reaches its maximum value, then so does $\ln \Omega_{AB}$. From the previous result, we guess that the "equilibrium" state is reached when:

$$\frac{q_A}{N_A} \approx \frac{q_B}{N_B} \Rightarrow q_A = q_B = 500$$

q_A	$\ln \Omega_A$	q_B	$\ln \Omega_B$	$\ln \Omega_{AB}$
0	0	1000	702.26	702.26
1	6.70	999	702	708.70
2	12.02	998	701.74	713.76
...	...	...	...	...
498	528.30	502	530.18	1058.48
499	528.78	501	529.72	1058.50
500	529.25	500	529.25	1058.50
501	529.72	499	528.28	1058.50
502	530.18	498	528.30	1058.48
...	...	...	...	...
998	701.74	2	12.02	713.76
999	702	1	6.70	708.70
1000	702.26	0	0	702.26

Determining the exact probability of each configuration is impossible, but we can easily figure out what the most probable configuration is inside system:

$$\Omega_{AB} \propto P(q_A, q_B)$$

The numerical results shows that our guess seems like correct. We consider the last case, in reality, q is often much larger than N, thanks to room temperature is very higher than absolute zero point.

Corollary 2.2.3.1 (Multiplicity of a Large Einstein Solid). *The multiplicity of a large Einstein solid at normal temperature ($q >> N$) is:*

$$\ln \Omega(N, q) = N + N \ln \frac{q}{N} \Rightarrow \Omega(N, q) = \left(\frac{eq}{N}\right)^N$$

Proof.

$$\begin{aligned}
\ln \Omega(N, q) &= (q + N) \ln (q + N) - q \ln q - N \ln N \\
&= (q + N) \ln \left[q \left(\frac{N}{q} + 1\right)\right] - q \ln q - N \ln N \\
&\approx (q + N) \left(\ln q + \frac{N}{q}\right) - q \ln q - N \ln N \\
&= N \ln \frac{q}{N} + N + \frac{N^2}{q} \\
&\approx N \ln \frac{q}{N} + N
\end{aligned}$$

We can see that:

$$\Omega(N, q) = \left(\frac{eq}{N}\right)^N$$

□

Suppose $N_B = 2N_A = 2.10^2$ and $q = q_A + q_B = 10^6$. Similarly, we guess that the "equilibrium state" is reached when:

$$\frac{q_A}{N_A} = \frac{q_B}{N_B} \Rightarrow q_A = 3.33.10^5,\ q_B = 6.67.10^5$$

q_A	$\ln\Omega_A$	q_B	$\ln\Omega_B$	$\ln\Omega_{AB}$
0	0	10^6	2042.66	2042.66
...	...	...	...	...
$3.20.10^5$	907.09	$6.80.10^5$	1826.30	2733.39
$3.33.10^5$	911.07	$6.67.10^5$	1822.44	2733.51
$3.40.10^5$	913.15	$6.60.10^5$	1822.33	2733.48
...	...	...	...	...
10^6	1021.03	0	0	1021.03

Example 2.2.1. What is the probability of event $E_1 : (q_A = 3.33.10^5, q_B = 6.67.10^5)$ occurring greater than the probability of event $E_2 : (q_A = 0, q_B = 10^6)$ occurring?

$$\frac{P(E_1)}{P(E_2)} = \frac{\Omega_1}{\Omega_2} = \frac{e^{2733.51}}{e^{2042.66}} \approx e^{690} \approx 10^{300} \text{ (times)}$$

So, there is no way E_2 could happen, compared to E_1.

After considering 3 simple cases of Einstein solids, you can see that equilibrium state is reached when energy is evenly distributed[1] within solids A and B. If we assume the initial internal energy of solid A is $q_A = 0$ and solid B is $q_B = q$, then very quickly the energy is **well-distributed** due to multiplicities. Solid A receives the energy from solid B, and solid B transfers its energy to solid A; until they both reach the equilibrium state. Since internal energy is directly proportional to temperature, heat will be transferred from the higher temperature object (solid B) to the lower temperature object (solid A). This is a very convincing answer for our previous question from previous chapter!

Theorem 2.2.4 (The Second Law of Thermodynamics)**.** *Multiplicity tends to increase; and any large system in equilibrium will be found in the macrostate with the greatest multiplicity.*

Corollary 2.2.4.1 (Heat flowing direction)**.** ***Statistically****, heat flows from the higher temperature object to the lower temperature object, and never in reverse.*

As the examples above have shown, when (q, N) are very large numbers, it is more convenient to deal with $\ln\Omega$[2] than Ω itself. Now we define a new quantity, called **entropy**, to scale the multiplicity $\ln\Omega$ down, and assign it with a physical unit.

Definition 2.2.1 (Entropy Definition)**.** *Entropy is defined as:*

$$S = k_B \ln\Omega \quad (J/K)$$

[1] I still can not prove the general result mathematically yet, as it is quite difficult. You could try proving the local maxima of Ω_{AB} is equilibrium state value:

$$\frac{q_A}{N_A} = \frac{q_B}{N_B}$$

A special case where $q >> N$ and $N_A = N_B = N$ is proved in section 2.4, "An Introduction to Thermal Physics", Daniel V. Schroeder.

[2] From now on, Ω is denoted as the short form for $\Omega(N, q)$.

In this chapter, we still do not explain why k_B is chosen to be scale factor yet, or physical meaning of the quantity S. Keep everything as simple as possible, and consider the quantity **entropy** as another representation of **multiplicity**. From Theorem 2.2.4, we can see that:

Corollary 2.2.4.2. *Entropy tends to increase.*

To conclude this section, we now develop some useful results in calculating entropy of a general Einstein solid.

2.2.1 N and q are relatively small

If both N and q values are small, you can calculate exactly entropy of an Einstein solid without using any approximation formula.

$$S = k_B \ln \Omega = k_B \ln \binom{N+q-1}{q}$$

2.2.2 N and q are large values

An Einstein solid at high-temperature

As we discussed above; if a solid is at high-temperature compared to absolute zero, then $q >> N$. We have proved the approximation formula:

$$\Omega \approx \left(\frac{eq}{N}\right)^N$$

Now we can conclude:

$$S = k_B \ln \Omega = k_B N \ln \frac{eq}{N}$$

An Einstein solid at low-temperature

At low temperature near absolute zero, intuitively $N >> q$, we have to develop another approximation formula for this case. Since this is not a typical case, we will not state any corollaries here.

$$\begin{aligned}
\ln \Omega &= (q+N)\ln(q+N) - q\ln q - N\ln N \\
&= (q+N)\ln\left[N\left(\frac{q}{N}+1\right)\right] - q\ln q - N\ln N \\
&\approx (q+N)\ln N + (q+N)\frac{q}{N} - q\ln q - N\ln N \\
&= q\ln N + \frac{q^2}{N} + q - q\ln q \\
&\approx q\ln\frac{N}{q} + q = \ln\left(\frac{eN}{q}\right)^q
\end{aligned}$$

Finally, we see that:

$$S = k_B \ln \Omega = k_B q \ln \frac{eN}{q}$$

A general Einstein solid model

Now we are considering the final case, which both N and q are large value; and their difference is **not significant**. For example, assume that $N = 2.31.10^5$ and $q = 2.43.10^5$, obviously we must derive our approximation from scratch because the initial conditions $q >> N$ or $N >> q$ are incorrect.

$$\begin{aligned}\Omega &= \binom{N+q-1}{q} = \frac{(N+q-1)!}{q!(N-1)!} = \frac{N}{q+N}\frac{(q+N)!}{q!N!}\\ &= \frac{N}{q+N}\frac{\sqrt{2\pi(q+N)}\,[(q+N)/e]^{q+N}}{\sqrt{2\pi q}\,(q/e)^q\,\sqrt{2\pi N}(N/e)^N} \quad \text{(Stirling's approximation)}\\ &= \frac{\sqrt{N}}{\sqrt{2\pi}\sqrt{q(q+N)}}\left(\frac{q+N}{q}\right)^q\left(\frac{q+N}{N}\right)^N\\ &= \frac{\left(\frac{q+N}{q}\right)^q\left(\frac{q+N}{N}\right)^N}{\sqrt{\frac{2\pi q(q+N)}{N}}}\end{aligned}$$

For any large values of N and q, the entropy of Einstein solid is:

$$S = k_B \ln\Omega = k_B \ln\left[\frac{\left(\frac{q+N}{q}\right)^q\left(\frac{q+N}{N}\right)^N}{\sqrt{\frac{2\pi q(q+N)}{N}}}\right]$$

2.3 The Ideal Gas Model

2.3.1 Fourier Transform and Quantum Mechanics

Every energy signals in time domain can be analyzed in frequency domain through Fourier transform[1][2]:

$$X(\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt = \mathcal{F}(x(t))$$

Similarly, spectrum in frequency domain can also be reconstructed to signal in time domain through inverse Fourier transform:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(\omega)e^{j\omega t}d\omega = \mathcal{F}^{-1}(X(\omega))$$

Now let's build a bridge. Imagine a single gas particle inside a box of volume V; it moves from point A to point B. All trajectories are in "position space", and its momentum is in "momentum space". Loosely speaking, if you know the particle's path from A to B, then you can relatively know its movement. Similarly, the transformation between volume space and momentum space can also be described as:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\psi(x)\exp\left(\frac{-jpx}{\hbar}\right)dx$$

[1] I think the most convenient way to represent the Fourier transform is complex exponential form, which I learned from Signals and Systems course in my university. You should refer to Chapter 3, "Signals and Systems", Alan V. Oppenheim for the full proof of these formulas.

[2] If this is your first time encountering the Fourier transform, I highly recommend watching this video to grasp the basic idea behind it.

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \phi(p) \exp\left(\frac{jpx}{\hbar}\right) dp$$

Let's take a look at these 2 formulas, you can see many similarities here. First, both frequency domain and momentum space are imaginary; conversely, both time domain and position space are real. Second, neglecting all constants, the Fourier transform and "position-momentum" transform are consistent. These quantities can be loosely related in physical meaning like this:

$$x(t) \text{ (Time domain)} \leftrightarrow \psi(x) \text{ (Position space)}$$

$$X(\omega) \text{ (Frequency domain)} \leftrightarrow \phi(p) \text{ (Momentum space)}$$

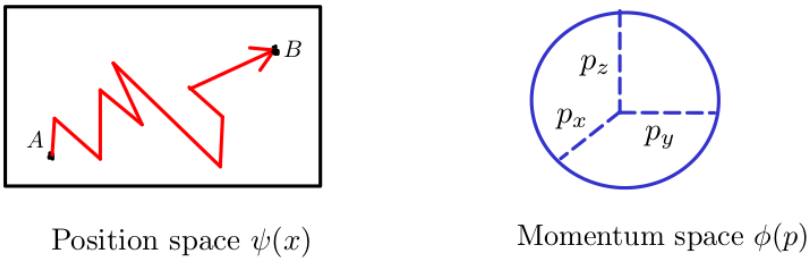


Figure 2.4: Position and momentum space

Why do I know the momentum space of a single gas particle is a sphere? Generally, the internal energy of a system is U, we have:

$$U = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2)$$

Equivalently, it yields:

$$p_x^2 + p_y^2 + p_z^2 = 2mU \text{ (Sphere equation)}$$

Notice that our momentum space $\phi(p)$ is just **the surface area** of the sphere. Now we move on to Heisenberg's uncertainty principle. Let's take a look at 2 signals and their Fourier representations.

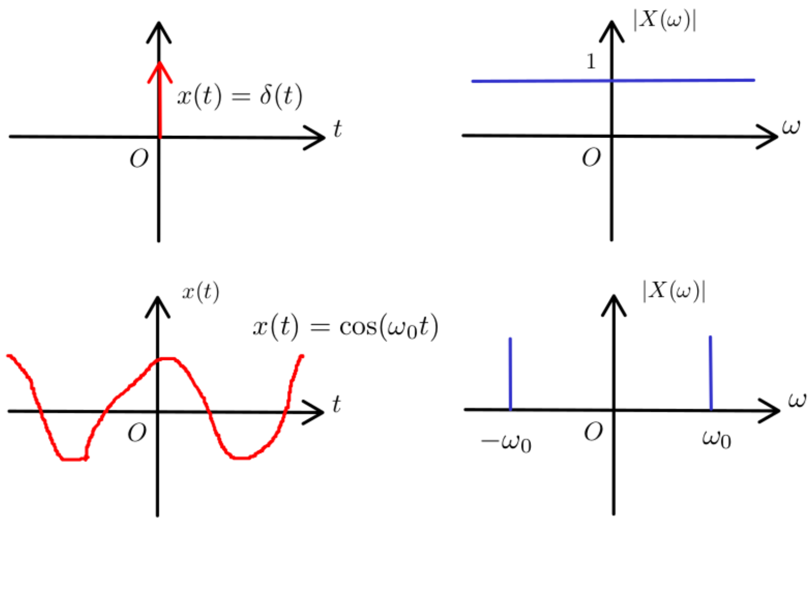


Figure 2.5: $\delta(t)$ and $\cos(\omega_0 t)$

The Dirac delta $\delta(t)$ signal focuses only on the origin in the time domain, but its spectrum spreads out entirely in the frequency domain. The cosine $\cos(\omega_0 t)$ signal spreads out over the time domain; but its spectrum focuses only on the frequency values $\omega = \omega_0$ and $\omega = -\omega_0$.

The more information you observe in the time domain, the less information you know in the frequency domain; and vice versa.

Using the same analogy, for simplicity, you can qualitatively think Heisenberg's uncertainty principle as:

The less spread out the wavefunction is in position space, the more spread out it must be in momentum space; and vice versa.

This statement can be rewritten as:

$$(\Delta x)(\Delta p_x) \approx h$$

where Δx is the spread in x, Δp_x is the spread in p, and h is Planck's constant.

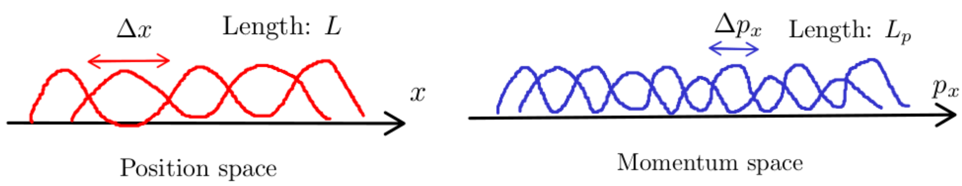


Figure 2.6: A number of position and momentum states for particle moving along x direction

The total number of distinct states on x direction is:

$$\frac{L}{\Delta x}\frac{L_p}{\Delta p_x} = \frac{LL_p}{h}$$

2.3.2 Entropy of Ideal Gas

In the previous subsection, I introduced a frame of quantum mechanics, that needed to calculate the entropy of ideal gas. Let's consider the simplest model, only one single ideal gas atom is inside the container. The system has internal energy U (mostly concentrates on the kinetic energy), and assume that the container's volume is V. How many microstates could the atom be in? We already know the number of available states on a single direction; using the same line of thinking, multiplicity on 3 directions x, y and z is:

$$\Omega_1 = \frac{VV_p}{h^3}$$

Notice that V_p is just **the surface area** of momentum space (in this case, it is a sphere). Add one more atom in our container, now what is the multiplicity of the system? Our momentum space now is more complicated:

$$p_{1x}^2 + p_{1y}^2 + p_{1z}^2 + p_{2x}^2 + p_{2y}^2 + p_{2z}^2 = 2mU$$

Obviously, this is not an ordinary surface equation; it defines the surface of a hypersphere in 6-dimensional momentum space.

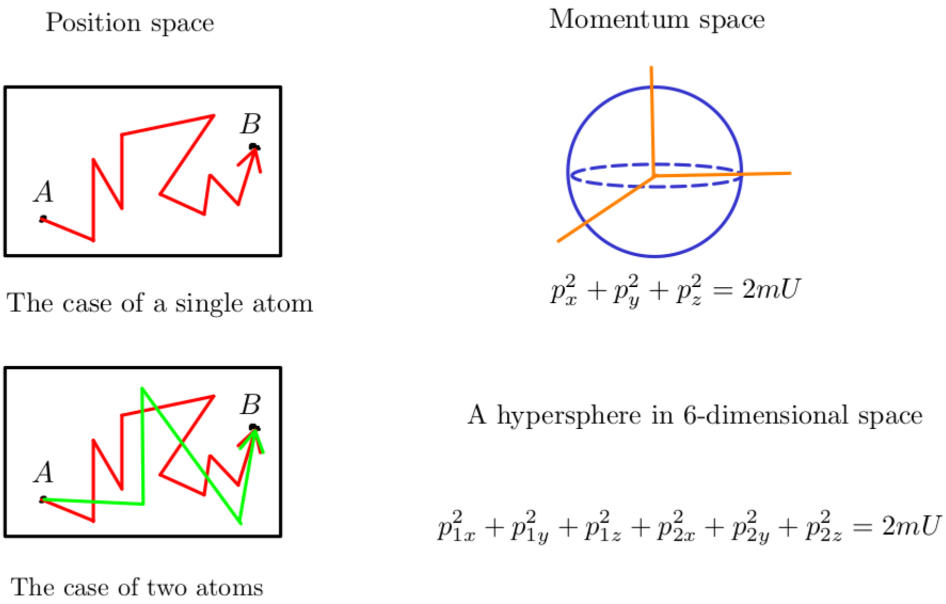


Figure 2.7: Multiplicity of a system containing 2 atoms

Similarly, we can see that:

$$\Omega_2 = \frac{V}{h^3}.\frac{V}{h^3} \times V_p = \frac{V^2}{h^6} \times V_p$$

But what is V_p? In the case of a single atom, V_p is the surface area of a sphere of radius $R = \sqrt{2mU}$. Is V_p still the "surface area" of a hypersphere in 6-dimensional space? The answer is yes, and we have a general formula for the "surface-area" of any d-dimensional hypersphere of radius r.

Theorem 2.3.1 ("Surface area" of d-dimensional hypersphere of radius r formula)**.**

$$A_d(r) = \frac{2\pi^{d/2}}{\left(\dfrac{d}{2} - 1\right)!} r^{d-1}$$

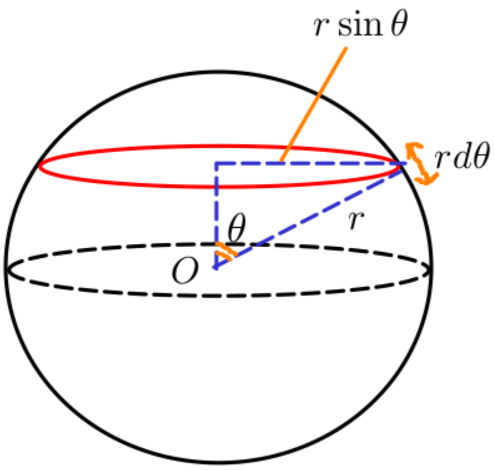
Proof. The simplest way to prove this result is to use induction. Rewriting in the form of Gamma function yields:

$$A_d(r) = \frac{2\pi^{d/2}}{\Gamma\left(\dfrac{d}{2}\right)} r^{d-1}$$

After substituing $d = 1, 2, 3$, we obtain:

$$\begin{aligned}
A_1(r) &= 2 \quad \text{(2 points bounding the line segment)} \\
A_2(r) &= 2\pi r \quad \text{(Circumference of a circle)} \\
A_3(r) &= 4\pi r^2
\end{aligned}$$

We do not care about the trivial case $A_1(r)$, but rather with guessing a recurrence relationship through how $A_3(r)$ is established.



Each red ring has circumference of $A_2(r\sin\theta)$ and a thickness of $rd\theta$

Figure 2.8: Derivation of $A_3(r)$ formula

It is obvious from the drawing that we have:

$$A_3(r) = \int_0^{\pi} A_2(r\sin\theta) r d\theta = \int_0^{\pi} 2\pi r \sin\theta . r d\theta = 4\pi r^2$$

The recurrence relation is pretty clear, we try to prove it is still true even $d > 3$.

$$A_d(r) = \int_0^{\pi} A_{d-1}(r\sin\theta) r d\theta$$

Assume that our formula still holds true up to $d-1$ dimension:

$$A_{d-1}(r) = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\dfrac{d-1}{2}\right)} r^{d-2}$$

Plug the function above into our recurrence relation:

$$A_d(r) = \int_0^{\pi} A_{d-1}(r\sin\theta) r d\theta = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\dfrac{d-1}{2}\right)} r^{d-1} \int_0^{\pi} \sin^{d-2}\theta d\theta$$

After separating the constant, now only the Wallis's integral left. To complete our proof, we have to show that:

$$\int_0^{\pi} \sin^{d-2}\theta d\theta = \frac{\sqrt{\pi}\Gamma\left(\dfrac{d-1}{2}\right)}{\Gamma\left(\dfrac{d}{2}\right)}$$

Setting $d-2 = n$, we reduce our equation to:

$$\int_0^{\pi} \sin^{n}\theta d\theta = \frac{\sqrt{\pi}\Gamma\left(\dfrac{n}{2} + \dfrac{1}{2}\right)}{\Gamma\left(\dfrac{n}{2} + 1\right)}$$

Before directly proving, let me recall the recursive property of Wallis's integral:

$$\begin{aligned}
W_n &= \int_0^\pi \sin^n \theta d\theta \\
&= \int_0^\pi \sin^{n-2} \theta (1 - \cos^2 \theta) d\theta \\
&= \int_0^\pi \sin^{n-2} \theta - \int_0^\pi \sin^{n-2} \theta \cos^2 \theta d\theta \\
&= \int_0^\pi \sin^{n-2} \theta d\theta - \int_0^\pi \cos \theta d \left(\frac{\sin^{n-1} \theta}{n-1} \right) \\
&= \int_0^\pi \sin^{n-2} \theta d\theta - \frac{1}{n-1} \int_0^\pi \sin^n \theta d\theta \\
&= W_{n-2} - \frac{1}{n-1} W_n
\end{aligned}$$

Finally, we can conclude:

$$W_n = \frac{n-1}{n} W_{n-2}$$

If n is an even integer, then:

$$\begin{aligned}
W_n &= \frac{n-1}{n} . \frac{n-3}{n-2} \cdots \frac{1}{2} . W_0 = \frac{(n-1)!!}{n!!} \pi \\
\Gamma \left(\frac{n}{2} + \frac{1}{2} \right) &= \frac{n-1}{2} . \frac{n-3}{2} \cdots \frac{1}{2} . \Gamma \left(\frac{1}{2} \right) = \frac{(n-1)!!}{2^{n/2}} \sqrt{\pi} \\
\Gamma \left(\frac{n}{2} + 1 \right) &= \frac{n}{2} . \frac{n-2}{2} \cdots \frac{2}{2} . \Gamma(0) = \frac{n!!}{2^{n/2}} . 1
\end{aligned}$$

We completed our proof for $n = 2k$ case. If n is an odd integer, then:

$$\begin{aligned}
W_n &= \frac{n-1}{n} . \frac{n-3}{n-2} \cdots \frac{2}{2} . W_1 = \frac{(n-1)!!}{n!!} . 2 \\
\Gamma \left(\frac{n}{2} + \frac{1}{2} \right) &= \frac{n-1}{2} . \frac{n-3}{2} \cdots \frac{2}{2} . \Gamma(0) = \frac{(n-1)!!}{2^{(n-1)/2}} . 1 \\
\Gamma \left(\frac{n}{2} + 1 \right) &= \frac{n}{2} . \frac{n-2}{2} \cdots \frac{1}{2} . \Gamma \left(\frac{1}{2} \right) = \frac{n!!}{2^{(n+1)/2}} \sqrt{\pi}
\end{aligned}$$

We completed our proof for $n = 2k + 1$ case. □

Back to our problem, now we obtain the multiplicity function for a system containing only 2 ideal gas atoms:

$$\Omega_2 = \frac{V^2}{h^6} \times V_p = \frac{V^2}{h^6} \times A_6(\sqrt{2mU})$$

But notice that 2 atoms are **indistinguishable**, it means they can be interchangeable:

$$\Omega_2 = \frac{1}{2!} . \frac{V^2}{h^6} \times V_p = \frac{1}{2!} . \frac{V^2}{h^6} \times A_6(\sqrt{2mU})$$

Generally, for a system containing N atoms, the multiplicity of a system is:

$$\Omega_N = \frac{1}{N!} . \frac{V^N}{h^{3N}} \times A_{3N}(\sqrt{2mU}) \approx \frac{1}{N!} . \frac{V^N}{h^{3N}} . \frac{\pi^{3N/2}}{(3N/2)!} (2mU)^{3N/2}$$

Theorem 2.3.2 (The Multiplicity of Idea Gas)**.** *The multiplicity of a system containing N ideal gas **particles** is:*

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{\pi^{3N/2}}{(3N/2)!}(\sqrt{2m})^{3N}U^{\boldsymbol{Nf}/2} = f(N)V^N U^{\boldsymbol{Nf}/2}$$

where f is the total number degrees of freedom of a single particle.

For monatomic gas case (as we discussed above) like He, f is equal 3; but f is equal 5 in case of H_2 or N_2.

Theorem 2.3.3 (The Entropy of Ideal Gas)**.** *The total entropy of a system containing N ideal gas atoms is:*

$$S = k_B \ln \Omega_N \approx N k_B \left[\ln\left(\frac{V}{N}\left(\frac{4\pi mU}{3Nh^2}\right)^{3/2}\right) + \frac{5}{2}\right]$$

This equation can be called as Sackur-Tetrode equation.

Proof.

$$\begin{aligned}
\ln \Omega_N &= -\ln N! + N\ln V - N\ln h^3 + \frac{3N}{2}\ln\pi - \ln\left(\frac{3N}{2}\right)! + \frac{3N}{2}\ln(2mU) \\
&\approx -N\ln N + N + N\ln V - N\ln h^3 + \frac{3N}{2}\ln\pi - \frac{3N}{2}\ln\frac{3N}{2} + \frac{3N}{2} + \frac{3N}{2}\ln(2mU) \\
&= N\left[\frac{5}{2} + \ln\left(\frac{V}{N}\frac{\pi^{3/2}}{h^3}\frac{(2mU)^{3/2}}{(3N/2)^{3/2}}\right)\right] \\
&= N\left[\frac{5}{2} + \ln\left(\frac{V}{N}\left(\frac{4\pi mU}{3Nh^2}\right)^{3/2}\right)\right]
\end{aligned}$$

We obtain:

$$S = k_B \ln \Omega_N \approx N k_B \left[\frac{5}{2} + \ln\left(\frac{V}{N}\left(\frac{4\pi mU}{3Nh^2}\right)^{3/2}\right)\right]$$

□

2.4 Entropy of Interacting Systems

From the previous section, if you look carefully at our calculated tables, you will see the tendency of **increasing entropy** in case of 2 Einstein solids interacting system.[1]

[1] I listed the three main formulas for entropy in Einstein solid model in subsection 2.2.1 and 2.2.2. For a deeper understanding of energy flow, you should try doing the numerical examples of calculations yourself to see it more clearly.

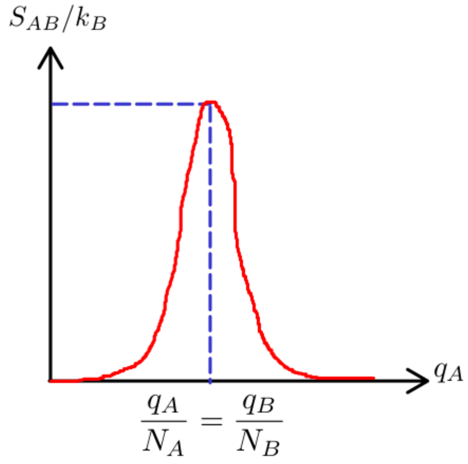


Figure 2.9: The tendency of increasing entropy graph

Now I will prove that this graph has only one local maxima in case of our system is at a high temperature, that is the most common and realistic case. The low temperature case can also be proven by the same method here[1]. We already knew that:

$$S_{AB} = S_A + S_B = k_B \ln \Omega_A + k_B \ln \Omega_B = k_B \left(N_A \ln \frac{eq_A}{N_A} + N_B \ln \frac{eq_B}{N_B} \right)$$

We will show that this function $S_{AB}(q_A, q_B)$ has only one local maxima by using two methods: examining single variable function and Lagrange multiplier. Let's begin with the easier one:

$$\begin{aligned} S_{AB} &= k_B \left(N_A \ln \frac{e}{N_A} + N_B \ln \frac{e}{N_B} \right) + k_B \left(N_A \ln q_A + N_B \ln q_B \right) \\ &= \text{constant} + k_B \left(N_A \ln q_A + N_B \ln q_B \right) \\ &= \text{constant} + k_B \left(N_A \ln q_A + N_B \ln (q - q_A) \right) \\ &= \text{constant} + k_B . f(q_A) \end{aligned}$$

Taking the first derivative of $f(q_A)$, and applying Fermat's theorem yields:

$$\frac{df(q_A)}{dq_A} = k_B \left(\frac{N_A}{q_A} - \frac{N_B}{q - q_A} \right) = 0 \Leftrightarrow N_A q_B = q_A N_B \Leftrightarrow \frac{q_A}{N_A} = \frac{q_B}{N_B}$$

Our function has exactly one critical point, but we do not know it is maxima or minima yet. Taking the second derivative and checking its sign may not be a good choice, since it is not so easy. We notice since $0 < q_A < q$, it is very easy to see that:

$$\lim_{q_A \to 0^+} f(q_A) = \lim_{q_A \to q^-} f(q_A) = -\infty$$

Drawing variation table of this function, we obtain:

[1]As I said before, I still can not prove for the general case.

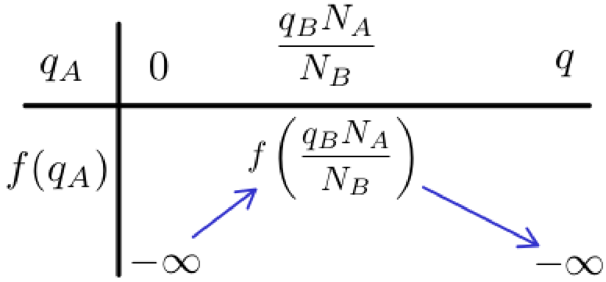


Figure 2.10: Variation table of $f(q_A)$

Now we conclude that $q_A = \dfrac{q_B N_A}{N_B}$ is the only local maximum. We move to the second method, applying Lagrange multiplier with initial constrain $q = q_A + q_B$:

$$\begin{cases} f(q_A, q_B) = N_A \ln q_A + N_B \ln q_B \\ g(q_A, q_B) = q_A + q_B = q \end{cases}$$

With the aid of Lagrange multiplier λ, we form a system of equations as follows:

$$\begin{cases} \nabla f(q_A, q_B) = \lambda \nabla g(q_A, q_B) \\ g(q_A, q_B) = q_A + q_B = q \end{cases} \quad \Leftrightarrow \begin{cases} \dfrac{N_A}{q_A} = \lambda \\ \dfrac{N_B}{q_B} = \lambda \\ q_A + q_B = q \end{cases}$$

Again, we also conclude that $q_A = \dfrac{q_B N_A}{N_B}$ is the only local maximum because the system of equations above has only one solution.
Now let's take a look at another interacting system of 2 ideal gases A and B; for simplicity, our gases are both **monoatomic**, due to Theorem 2.3.2, each atom has 3 degrees of freedom:

$$\Omega = f(N) V^N U^{3N/2}$$

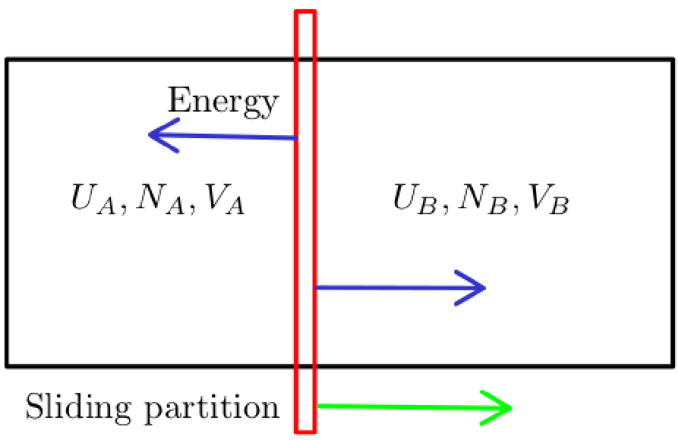


Figure 2.11: Energy can pass through partition, but not particles.

The total entropy of our system is:

$$\begin{aligned}
S_{AB} &= k_B \ln \Omega_{AB} = k_B \left(\ln \Omega_A + \ln \Omega_B \right) \\
&= \text{constant} + k_B \left(N_A \ln V_A + \frac{3N_A}{2} \ln U_A + N_B \ln V_B + \frac{3N_B}{2} \ln U_B \right) \\
&= \text{constant} + k_B \left(N_A \ln V_A + N_B \ln V_B \right) + k_B \left(\frac{3N_A}{2} \ln U_A + \frac{3N_B}{2} \ln U_B \right)
\end{aligned}$$

Precisely determining the critical point is not so easy, because we have to use 2 Lagrange multipliers corresponding to 2 fixed conditions: $N = N_A + N_B$ and $U = U_A + U_B$. However, we can quantitatively the graph of U_A, V_A, S_{AB}/k_B from previous conclusion.

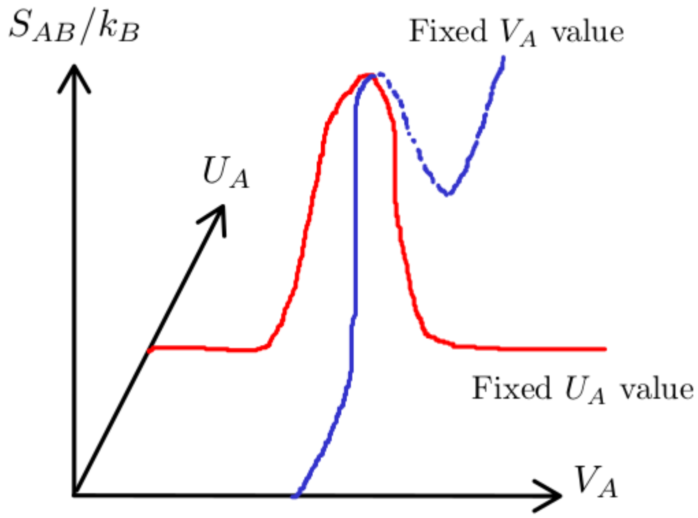


Figure 2.12: Entropy of a system of 2 ideal gases, with sliding partition

The red curve represents the S_{AB}/k_B graph (in form of our previous result has been proved) while holding an arbitrary U_A value fixed; and so does the blue curve. Now, the commonality between these two graphs can be clearly seen; both have **only** a single extreme peak, and regardless of the initial state, entropy always increase; or in other words, the system always approaches its most probable state. As you can see here, since entropy is always increasing, we do not need to worry so much about the initial S_i or final S_f state, but rather about **difference** $\Delta S = S_f - S_i$ between them.

Theorem 2.4.1 (Entropy is increasing)**.**

$$\Delta S \geq 0$$

If any process can hold $\Delta S = 0$, that means $S_i = S_f$, we call it **reversible process**; and if $\Delta S > 0$, or equivalent to $S_f > S_i$, we call it **irreversible process**. Keep these words in mind, we will return to them at the end of next chapter, when we work with some well-known thermodynamics cycles; they will be revisited as a result of the second law. To conclude this chapter, I will introduce you to two classic irreversible processes: **free expansion** and **gas mixing**.

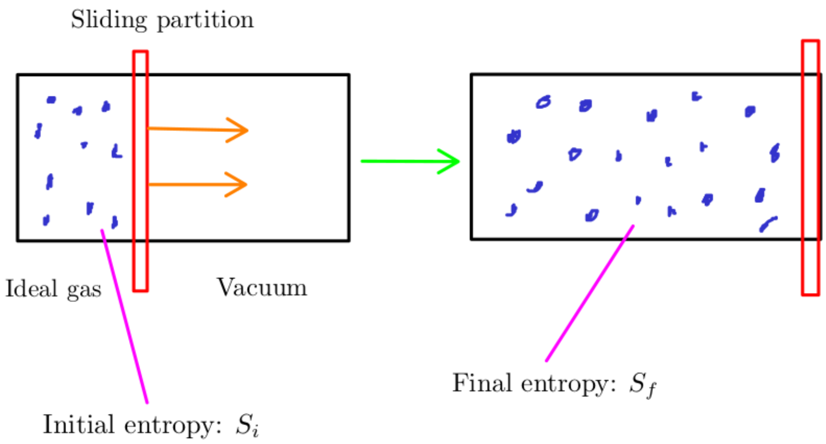


Figure 2.13: Free expansion phenomenon

Due to Theorem 2.3.3, the entropy difference is:

$$\Delta S = Nk_B \ln \frac{V_f}{V_i}$$

As you can see here $V_f > V_i$ due gas expansion to fill the container, even it does not do any work or receive any external heat:

$$\Delta U = Q + W = 0 + 0 = 0$$

This empirical result makes sense because it satisfies our law: $\Delta S \geq 0$. Now suppose you have equal moles of two ideal gases A and B, divide them into two equal parts inside a container. What happens if you pull the partition out?

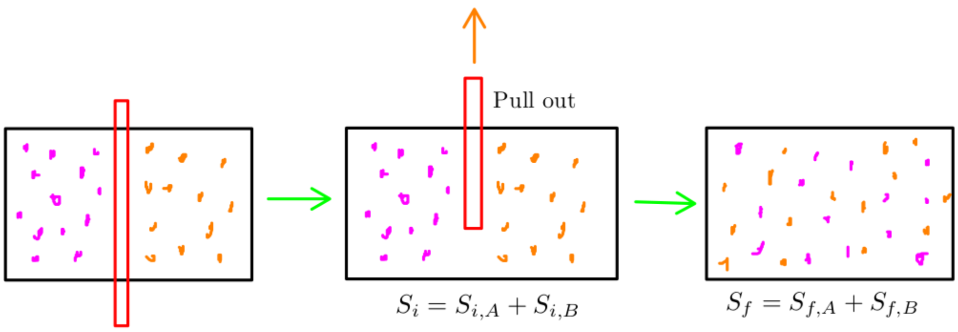


Figure 2.14: Gas mixing phenomenon

For simplicity, I assume that:

$$V_{i,a} = V_{i,b} = \frac{V}{2}$$

We can see that:

$$\Delta S = \Delta S_A + \Delta S_B = N_A k_B \ln \frac{V_f}{V_i} + N_B k_B \ln \frac{V_f}{V_i} = 2Nk_B \ln 2$$

Again, entropy increased in the process of mixing gas. We have successfully applied the second law to explain the expansion of gases inside a container. But how about the shrinking case?

Perhaps instead of mixing or expanding, these gases particles just stay still in one place, or even shrink. The answer is these phenomenons could never happen, because they **violate the second law of thermodynamics**.

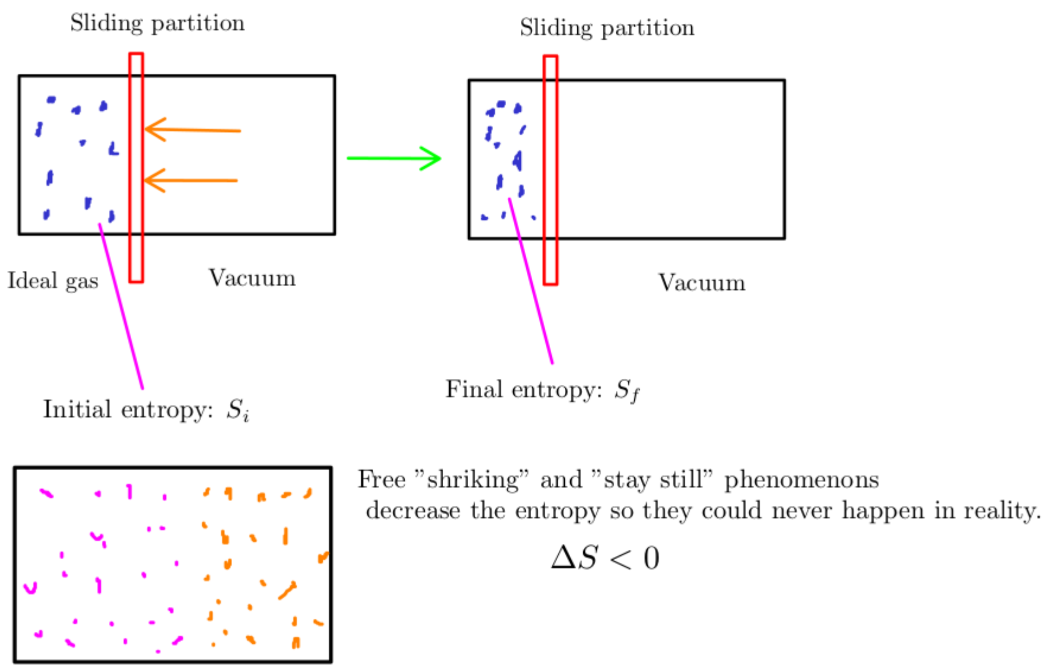


Figure 2.15: The cases of violating the second law of thermodynamics

Chapter 3

The Third Law of Thermodynamics

Nobody knows what entropy really is, so in any discussion you will always have an advantage.[1]

3.1 The Meaning of Entropy

From the previous chapter, we have been working with the term "entropy" in various situations: solids, ideal gases; but what exactly is entropy? Unlike other quantities such as volume (V), pressure (P), number of particles (N),...; it is not so easy to imagine an abstract and "invisible" thing like entropy (S). In this section, we discuss an intuitive way of understanding entropy, without relying on formulas. Let's forget everything you have learned so far, and decorate a blank wall with a few pictures. Now I give you 5 landscape pictures, and your mission is sticking them on my wall.

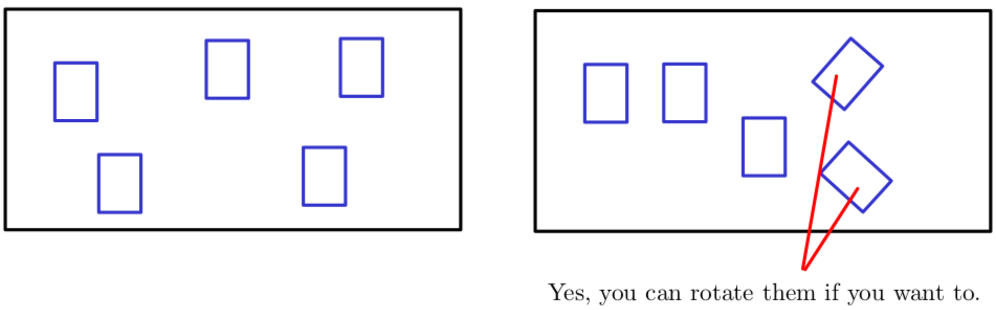


Figure 3.1: 2 possible picture arrangements

There are many possible ways of sticking pictures; I can not draw and count all of them here, but the number of possible arrangements is **not infinity**. You can think of entropy as the number of ways you can arrange the pictures on the wall. The more arrangements, the higher the entropy of our system (just consists a wall with five pictures). Obviously, if our wall is larger, we are more freely to stick our pictures wherever we want; and the entropy of our new system (again, larger wall with five picture) is greater than the smaller one.

[1]A quote from John von Neumann, when he discussed the idea of entropy with Claude Shannon. I have heard this term is also used in information theory, but that is not our concern here.

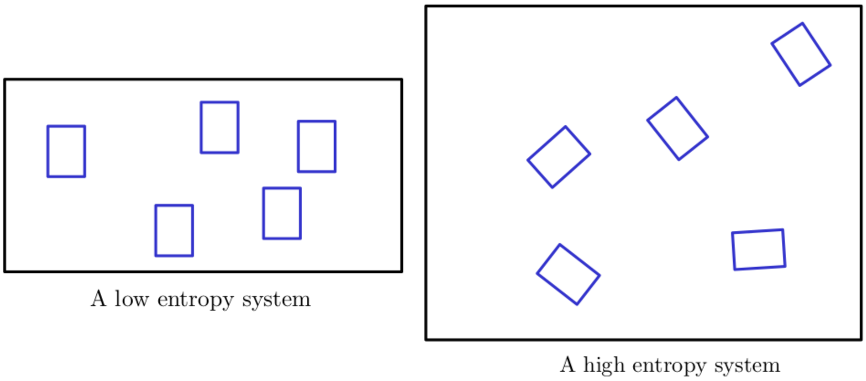


Figure 3.2: Large system has more entropy than the small system

You can also increase the total entropy by sticking more and more pictures on the wall too. But now, do you see that if there are many different ways to arrange pictures, the wall will become even more cluttered? By the same logic, the higher the entropy, the more chaotic the system becomes? "Why is more cluttered? Bro, no matter the size of the wall, I always arrange everything parallel, like all pictures are in one row.". Yes, nothing is wrong; however, we need to note a subtle difference between **intentionally** arranging pictures in one row, and **randomly** sticking them on our wall. Apparently, when we concern how many possible arrangements, we are dealing with the **randomness**. Randomly sticking pictures on a small wall is less "chaotic" (or "messy") than on a larger one; since there are fewer ways to do so.

Entropy is a quantity that describes how chaotic (or messy) a system is.

Now imagine you are sticking pictures on your wall A, nearby is my wall B. I ask if you could decorate my wall too, so our big wall would be so nice; and I assume that you agree.

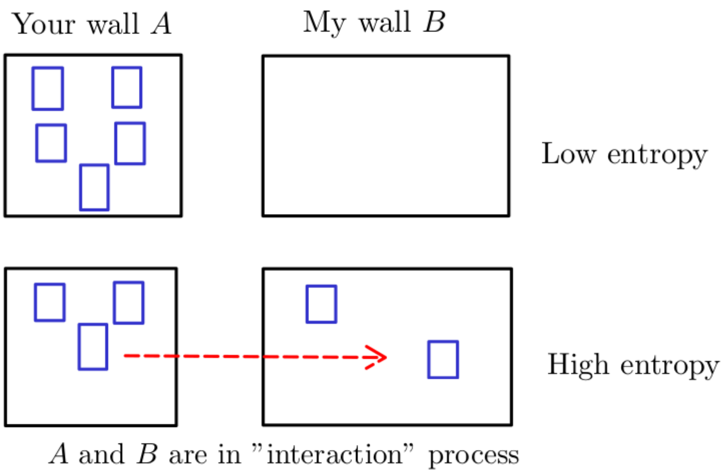


Figure 3.3: The total entropy of wall $A + B$ system

When you start moving pictures to my wall and sticking them on, we can see that our wall are "in contact" and "transferring" pictures with each other. This "interaction" process increases the total entropy of our system[1].

The interaction process increases the total entropy of a system.

Inductively, we imagine the universe as a giant closed system, with many particles inside; and they are always exchanging "something"[2] with each other, we can firmly state:

The total entropy of the universe is increasing.

An ice cube is more "organized" than a cup of water, and vice versa. In the middle of winter, water is frozen. Does this phenomenon violate the second law of thermodynamics? Probably, because the liquid is now condensing (or "organizing") into a solid block of ice; obviously its entropy decreases!

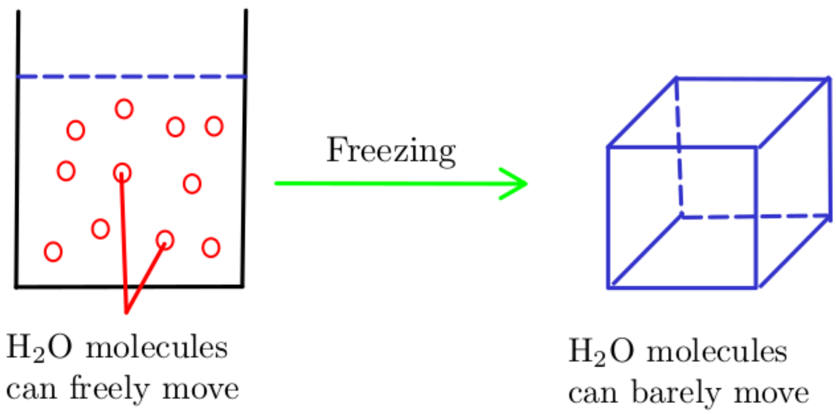


Figure 3.4: Water is frozen

Surprisingly, the answer is no. The water is cooled down since it **contacts** with the outside environment; they are **interacting**. Because the water's temperature is much higher than surrounding; so it gives up some of its energy, while the environment receives them. A decrease in energy leads to a decrease in entropy (particles become less mobile, thus making their movement more difficult), and vice versa. From the law of conservation of energy:

$$\textbf{Energy released by water} = \textbf{Energy received by environment}$$

But:

$$\textbf{Entropy "released" by water} \neq \textbf{Entropy "received" by environment}^3$$

Entropy is closely related to energy, but they are not the same, and we do not have any law of conservation of entropy; so we have to be careful here. This result below may surprise you:

$$\textbf{Entropy "released" by water} < \textbf{Entropy "received" by environment}$$

So the total entropy of our system (a cup of water and the outside environment) increases, because it gains more than losses.

[1]This conclusion is true, both in our common sense and in our calculations. We proved several special cases in the Chapter 2.

[2]I do not know if the completely isolated object exists or not; its existence violates our laws of thermodynamics.

[3]I use the double bracket here to emphasize the fact that "entropy" is not transferred in the way of energy or other physics quantities.

3.2 The Meaning of Temperature

In the conventional sense, a cup of boiling water has much higher temperature than its surrounding, so it will transfer some of its energy to surrounding environment (and conversely, the environment also receives that heat), according to the zeroth law of thermodynamics. As we discussed in the previous section:

Entropy is directly related to heat.

And obviously, also by the law:

The amount of heat transferred between objects is directly linked to the temperature difference.

The temperature itself is not so important than the **difference**. If our system only consists objects at same temperature, nothing is exchanged, everything is in equilibrium state. Only when $\Delta T > 0$, some heat Q is transferred, from the high temperature objects to the lower; which leads to the **total entropy of entire system** slowly **increases**. In our thinking line, we can imagine:

Entropy is something that deeply related not only heat, but also the temperature difference too.

In this section, we develop the connection among three physical quantities: $(\textbf{heat}, \textbf{energy}, \textbf{entropy})$. Let's begin with a very classical problem: consider a system consists two Einstein solids A ($N_A = 40$) and B ($N_B = 60$), and the total energy is $q = 100$. Using the same calculating techniques from Chapter 2, we obtain some numerical values:

q_A	$\ln \Omega_A$	q_B	$\ln \Omega_B$	S_{AB}/k_B
0	0	100	105.85	105.85
1	4.70	99	105.37	110.07
...	...	...	...	...
20	38.19	80	95.60	133.79
21	39.27	79	95.04	134.31
22	40.32	78	97.60	134.92
...	...	...	...	...
39	54.75	61	83.86	138.61
40	55.45	60	83.17	138.62
41	56.13	59	82.48	138.61
...	...	...	...	...
100	83.75	0	0	83.75

Plot our data into a simple graph:

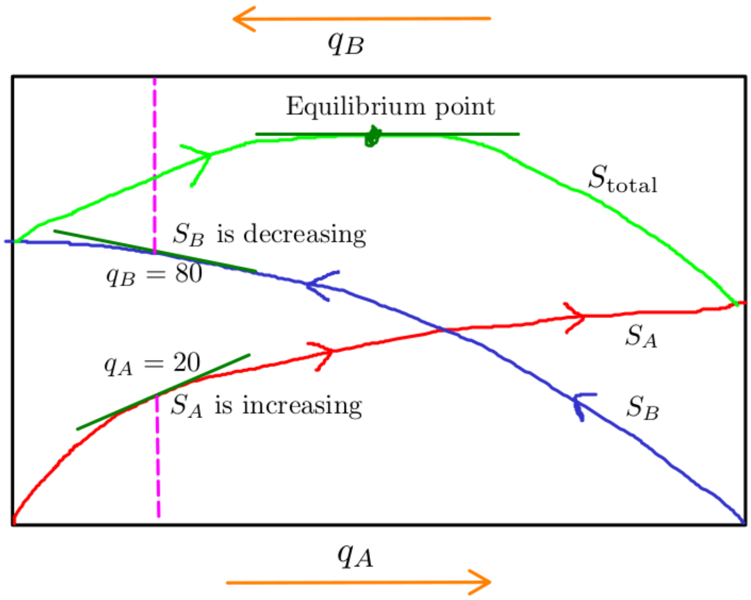


Figure 3.5: A simple graph of the entropy

Looking at table and graph, we find out some interesting observations here. I randomly pick an interval; q_A is increasing 2 units, from $q_A = 20$ to $q_A = 22$, its entropy is also increasing $\Delta S_A = 40.32 - 38.19 = 2.13$ (k_B); meanwhile S_B is decreasing $\Delta S_B = 94.6 - 95.6 = -1$ (k_B). We can see that:

$$\Delta S_{\text{total}} = \Delta S_A + \Delta S_B = 2.13 + (-1) = 1.13 \ (k_B)$$

Again, the total entropy is increasing! Can I guess the slope of a tangent line at point $q_A = 20$ is **steeper** than point $q_B = 80$? Let's find out by doing some math here. Recall from our approximation formula:

$$S = k_B[(q+N)\ln(q+N) - q\ln q - N\ln N] \Rightarrow \left(\frac{\partial S}{\partial q}\right)_N = k_B\left[\ln\left(1+\frac{N}{q}\right)\right]$$

Substituting our values yields:

$$q_A = 20, N_A = 40 \Rightarrow \frac{\partial S_A}{\partial q_A} = 1.09k_B$$

$$q_B = 80, N_B = 60 \Rightarrow \frac{\partial S_B}{\partial q_B} = 0.55k_B$$

Here is a question, why is ΔS_B negative, but it still has positive slope? The answer is pretty simple, our graph perspective is from right to left, but calculation convention is from left to right. If the initial state of our system is $(q_A = 0, q_B = 100)$, then the total entropy is slowly increasing, until it reaches its **maximum value**, and stays still at equilibrium point. There is no chance for our system could fall out of its equilibrium state anymore, after it reached [1].

[1]Or more precisely, no matter where it begins, the probability of equilibrium state occurring is always the highest. I recommend you should verify this fact by calculating the ratio:

$$\frac{P(q_A = 40, q_B = 60)}{P(q_A = 45, q_B = 55)}$$

Definition 3.2.1 (Equilibrium Definition)**.** *Equilibrium state is the state which entropy reaches its* ***maximum value****, or equivalently:*

$$\Delta S = 0$$

According the zeroth law of thermodynamics, the temperature of solid A and B are equal if and only if they are in equilibrium, that means S_{total} is now maximum. Obviously, equilibrium point is the local maxima, so its tangent line is now **straight**. We take the first derivative with respect to q_A :

$$\frac{\partial S_{\text{total}}}{\partial q_A} = 0 \Rightarrow \frac{\partial (S_A + S_B)}{\partial q_A} = 0 \Rightarrow \frac{\partial S_A}{\partial q_A} = -\frac{\partial S_B}{\partial q_A}$$

A subtle observation is that a decrease in q_A corresponds to an increase in q_B:

$$\frac{\partial S_A}{\partial q_A} = \frac{\partial S_B}{\partial q_B}$$

This equation makes perfect sense to our intuition, look at previous calculation: the slope of S_A graph is much steeper than S_B, then it gradually decreases, while the slope of S_B slowly increases until they are **equal**. That is an equilibrium moment. Now if our system is not in equilibrium, I assume:

$$\frac{\partial S_A}{\partial q_A} > \frac{\partial S_B}{\partial q_B}$$

To ensure the second law of thermodynamics, as you can see here, the entropy of solid A increases **faster** than the rate of entropy decrease of solid B. That means solid A receives energy released from solid B; and now we can conclude that **the temperature of solid A is lower than solid B.** Before introducing the canonical definition of temperature, let me define a new "energy" quantity unit ε[1]:

$$U = \varepsilon . q$$

Apparently, we can see that:

$$\frac{\partial S_A}{\partial q_A} = \frac{\partial S_B}{\partial q_B} \Leftrightarrow \frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$$

Definition 3.2.2 (The Third Definition of Temperature)**.** *The temperature is defined as:*

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V} \quad (1/K)$$

We now eventually understood why the unit of entropy is (J/K) (thanks to the unit of k_B constant). Plugging it to the definition above, the energy part (J) disappears, so only (1/K) left.

3.3 Proof of Equipartition Theorem

In this section, we will provide a simple proof of equipartition theorem (Theorem 1.5.1) in the two simplest cases: Einstein solid and ideal monatomic gases. Let's begin with the entropy formula of a solid at high temperature (Subsection 2.2.2):

$$S = k_B N \ln \frac{eq}{N} = k_B N \ln \frac{eU}{\varepsilon N} \Rightarrow \frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V} = \frac{k_B N}{U} \Rightarrow U = k_B N T$$

[1]The unit of ε is (eV), which is the kinetic energy of an electron that has been accelerated through a voltage different of 1V.

Is there any conflict here? Since we already know that any solid has 6 degrees of freedom, its internal energy must be:

$$U = \frac{f}{2}k_B NT = 3k_B NT \quad \text{(Corollary 1.5.1.1)}$$

Here is the subtle observation, the key difference is how the quantity N defined:

$$U = k_B NT \quad (N \text{ is } \textbf{the number of chemical bonds inside solid})$$
$$U = 3k_B NT \quad (N \text{ is } \textbf{the number of particles inside solid})$$

Inside crystal lattice, each particle has 3 chemical bonds, so our proof is correct:

$$N_{\textbf{bonds}} = 3N_{\textbf{particles}}$$

For convention, we agree that N is **the number of chemical bonds inside Einstein solid**; so the heat capacity of solid at constant volume is:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{N,V} = k_B N$$

For ideal monatomic gas, from Theorem 2.3.3, take the partial derivative with respect to U, we obtain:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V} = \frac{3}{2}Nk_B\frac{1}{U} \Rightarrow U = \frac{3}{2}Nk_B T$$

Again, the equipartition theorem still holds true. Similarly, the heat capacity at constant volume is:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{N,V} = \frac{3}{2}Nk_B$$

3.4 The Third Law of Thermodynamics

From the definition of temperature (Definition 3.2.2):

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$$

An infinitesimally small change in entropy is equivalent to:

$$dS = \frac{dU}{T}$$

We assume that our system does not do any kind of works:

$$dS = \frac{dU}{T} = \frac{Q+W}{T} = \frac{Q}{T} = \frac{C_V dT}{T} \Rightarrow \Delta S = \int_{T_i}^{T_f} \frac{C_V dT}{T}$$

This seems like a very easy way; you can calculate the difference in entropy of any system with constant volume, but here is the tricky part. I set the initial temperature $T_i = 0$ (K):

$$\Delta S = S_f - S(0) = \int_0^{T_f} \frac{C_V dT}{T} = C_V.\lim_{t\to 0^+}(\ln T_f - \ln t)$$

This integral is divergence, and might be $\Delta S \to +\infty$, do you think that makes sense? The answer is no, there is no way entropy (or any other physical quantities) can be infinite. To ensure convergence, our mathematics indicates condition:

$$T \to 0 \text{ (K)} \Rightarrow C_V \to 0 \text{ (J/K)}$$

How about $S(0)$? Intuitively, we imagine the state where everything is "frozen", and due to equipartition theorem (Theorem 1.5.1), the internal energy is equal zero, leads to number of units energy is also zero; and there is **only one** configuration that satisfies $q = 0$.

$$S = k_B \ln \Omega = k_B \ln 1 = 0 \text{ (J/K)}$$

Theorem 3.4.1 (The Third Law of Thermodynamics)**.** *At absolute zero temperature a system settles into its unique lowest-energy state:*

$$\lim_{T \to 0} S = S(0) = 0 \textit{ (J/K)}$$

Theorem 3.4.2 (The Macroscopic View of Entropy)**.** *For any system, the difference in entropy can easily be determined as follows:*

$$dS = \frac{dU}{T} \Rightarrow \Delta S = \int_i^f \frac{dU}{T} \textit{ (J/K)}$$

Example 3.4.1. 100 grams of water is heated from 298K to 328K at **constant volume** condition; given the specific heat of water is $c_V = 4.13$ (kJ/kg.K).
The amount of increased entropy is:

$$\Delta S = \int_i^f \frac{dU}{T} = \int_{T_i}^{T_f} \frac{C_V dT}{T} = C_V . \ln \frac{T_f}{T_i} = m.c_V . \ln \frac{T_f}{T_i} = 0.1.4.13.0.096 = 0.0413 \text{ (kJ/K)}$$

100 grams of water at 328K (55°C) has:

$$\frac{\Omega_f}{\Omega_i} = \exp\left(\frac{\Delta S}{k_B}\right) = \exp\left(3.10^{24}\right) \text{ (times)}$$

more "chaotic" than at 298K (25°C).

3.5 Various Types of Equilibrium

3.5.1 Thermal Equilibrium

Now it is time to focus on the most common types of equilibrium in thermodynamics systems. We discussed thermal equilibrium in the Chapter 0 of this book, but I want to recall some important properties here, and we will solve a simple example directly related to this phenomenon.

Definition 3.5.1 (Definition of Thermal Equilibrium)**.** *Systems are in thermal equilibrium when **their temperatures** are the same.*

From Definition 3.2.2:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V}$$

Example 3.5.1. A 50 grams copper sphere at 100°C is dropped into a cup containing 100 grams of water at 25°C. Given $c_{V,\mathrm{H_2O}} = 4.13$ (kJ/kg.K) and $c_{P,\mathrm{Cu}} = 386$[1] (J/kg.K); show that the total entropy of system is increasing.

Using the same technique from the last example in Chapter 0, we obtain the equilibrium (final) temperature is $T_f = 301.34K$ (28.34°C).

$$\Delta S_{\mathrm{H_2O}} = \int_i^f \frac{dU}{T} = C_V.\int_{T_i}^{T_f} \frac{dT}{T} = m_{\mathrm{H_2O}}.c_{V,\mathrm{H_2O}}.\int_{T_i}^{T_f} \frac{dT}{T} = 0.1.4.13.10^3.\ln\frac{301.34}{298} = 4.6(\mathrm{J/K})$$

Here is the subtle part of this problem, how do we know ΔS_{Cu}? We only know its $c_{P,\mathrm{Cu}}$ value, and can not directly apply the formula above. When looking at our equation, I notice that I can remove something:

$$dU = Q + W = Q + P\Delta V = Q + P(\beta.V.\Delta T) = Q + P(\beta.\rho.m.\Delta T)^2 \approx Q$$

The difference in entropy is:

$$\Delta S_{\mathrm{Cu}} = \int_i^f \frac{dU}{T} = C_P\int_i^f \frac{dT}{T} = m_{\mathrm{Cu}}.c_{P,\mathrm{Cu}}\int_{T_i}^{T_f} \frac{dT}{T} = 0.05.386.\ln\frac{301.34}{373} = -4.1\ (\mathrm{J/K})$$

Finally, we now can conclude that the total entropy is increasing, as expected:

$$\Delta S = \Delta S_{\mathrm{H_2O}} + \Delta S_{\mathrm{Cu}} = 4.6 - 4.1 = +0.5\ (\mathrm{J/K})$$

3.5.2 Mechanical Equilibrium

Definition 3.5.2 (Definition of Mechanical Equilibrium)**.** *Systems are in mechanical equilibrium when **their pressures** are the same.*

The novelty in this definition is that we will redefine the concept of pressure in terms of temperature and entropy. Let's consider a simple model: two ideal gas A and B, divided by a sliding partition. They can exchange energy, but not particles. To make it easier to visualize, I have drawn Figure 2.11 and Figure 2.12. On the peak of S_{total}, you can see that both partial derivatives with respect to U_A and V_A are equal zeros.

$$\frac{\partial S_{\mathrm{total}}}{\partial U_A} = 0 \Rightarrow \frac{\partial S_A}{\partial U_A} = -\frac{\partial S_B}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$$
$$\frac{\partial S_{\mathrm{total}}}{\partial V_A} = 0 \Rightarrow \frac{\partial S_A}{\partial V_A} = -\frac{\partial S_B}{\partial V_A} = \frac{\partial S_B}{\partial V_B}$$

Now we form the conditions for mechanical equilibrium:

$$\begin{cases} \dfrac{\partial S_A}{\partial U_A} = \dfrac{\partial S_B}{\partial U_B} & \text{(Thermal Equilibrium)} \\ \dfrac{\partial S_A}{\partial V_A} = \dfrac{\partial S_B}{\partial V_B} & \end{cases}$$

Obviously, if a system is in mechanical equilibrium; it **must be in thermal equilibrium**.

[1] I could not find any detailed information about $c_{P,\mathrm{Cu}}$ value on the internet; but that is understandable because we usually measure c_V of liquids and gases, c_P of solids. Maintaining constant volume condition (for liquids and gases) and constant pressure (for solids) is much easier when conducting any type of measurement.

[2] β: coefficient of volume expansion (1/K), ρ: density (kg/m^3). The product result is small enough (you should verify yourself) to be neglected.

Definition 3.5.3 (The Second Definition of Pressure). *The pressure of any system is defined as:*

$$P = T\left(\frac{\partial S}{\partial V}\right)_{N,U} \qquad (Pa)$$

This definition looks very weird, but somehow it still makes perfect sense. Let me explain the way I understand it. Consider the ideal gas law:

$$PV = Nk_BT \Rightarrow P = \frac{Nk_BT}{V}$$

The product on the right side is **proportional** to the internal energy U, so I can rewrite our equation to:

$$P \propto \frac{U}{V} \Rightarrow P = \text{constant.}\left(\frac{\Delta U}{\Delta V}\right)$$

Look carefully to the units both sides, you can see the relationship between energy and volume is described by pressure quantity. Intuitively, if you increase the volume with fixed internal energy, pressure decreases; and similarly, if you hold the volume and increase internal energy (by adding more gas, or heat the system), pressure increases. According to the Theorem 3.4.2:

$$dU = TdS$$

Substituting to the previous equation, and choose T to be the constant:

$$P = T\left(\frac{\partial S}{\partial V}\right)$$

3.5.3 Diffusive Equilibrium

Before explaining what these terms are, let me introduce you to some examples of "diffusion" first: mixing tea and milk together, or gases mixing phenomenon (Figure 2.14). I often think that diffusion is closely related to mixing.

Definition 3.5.4 (Definition of Diffusion). *Diffusion is the net movement of anything generally from a region of higher concentration to a region of lower concentration.*

But you should note that these phenomenons are not examples of diffusion: mixing H_2 and Cl_2, or any kind of chemical reaction.

Definition 3.5.5 (Definition of Diffusive Equilibrium). *Systems are in diffusive equilibrium when* ***their chemical potentials*** *are the same.*

What is chemical potential? I will start with a simple container, contains 2 ideal gases A and B, separated by a partition. Then I pull out the partition, the gases start mixing together, and when they reach equilibrium state:

$$\frac{\partial S_{\text{total}}}{\partial U_A} = 0 \Rightarrow \frac{\partial S_A}{\partial U_A} = -\frac{\partial S_B}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$$
$$\frac{\partial S_{\text{total}}}{\partial N_A} = 0 \Rightarrow \frac{\partial S_A}{\partial N_A} = -\frac{\partial S_B}{\partial N_A} = \frac{\partial S_B}{\partial N_B}$$

Now we form the conditions for diffusive equilibrium:

$$\begin{cases} \dfrac{\partial S_A}{\partial U_A} = \dfrac{\partial S_B}{\partial U_B} & \text{(Thermal Equilibrium)} \\ \dfrac{\partial S_A}{\partial N_A} = \dfrac{\partial S_B}{\partial N_B} \end{cases}$$

Definition 3.5.6 (Definition of Chemical Potential)**.** *The chemical potential of any system is defined as:*

$$\mu = -T\left(\frac{\partial S}{\partial N}\right)_{U,V} \qquad (eV)$$

Particles tend to flow from the system with ***higher μ into the system with lower μ.***

A very strange quantity (again?), but it is very useful in describing how fluids flow. For example, if I mix cold tea and hot milk together, I can calculate their chemical potentials and know whether the milk tends to flow into the tea or in the reverse direction. I often relate it to electrical potential, both tell us the tendency to flow, from high to low potential. What does the negative sign do? Let's find out; first I define a new quantity u:

$$u = \frac{\partial U}{\partial N}$$

The quantity u describes the rate of change of U; if u is high, it means that U is increasing rapidly and particles **tends to flow into** the fluid we are considering. If I mix fluids A and B together, known that $u_A > u_B$, then I can conclude that the flow of particles is from B to A. There is nothing wrong here, except I feel a little but uncomfortable; perhaps I should add a negative sign to ensure our calculating direction is not reversed?

$$-u_B > -u_A \Leftrightarrow \text{ Flowing direction: } B \to A$$

My small modification works very well, I want to connect it with other quantities:

$$-u = -\frac{\partial U}{\partial N} = -T\left(\frac{\partial S}{\partial N}\right) = \mu$$

That is how I imagine the way of chemical potential μ was defined. But unlike electrical potential, μ_A and μ_B **constantly change**, until they are equal in equilibrium state.

3.5.4 The Thermodynamics Identity

For any changing in entropy in any thermodynamics process:

$$dS = \frac{dU}{T} + \frac{PdV}{T} - \frac{\mu dN}{T}$$

Theorem 3.5.1 (The Thermodynamics Identity)**.**

$$dS = \frac{dU}{T} + \frac{PdV}{T} - \frac{\mu dN}{T} \Leftrightarrow dU = TdS - PdV + \mu dN$$

This identity will play crucial role in later chapters.

3.6 Heat Engines and Refrigerators

[1] In the final section of this chapter, we will study two other versions of the second law of thermodynamics.

Definition 3.6.1 (Heat Engine)**.** *A heat engine is a device that extracts energy from its environment in the form of heat and does useful work.*

[1] This section is unrelated to the rest of this book; so you may omit or postpone it if you wish.

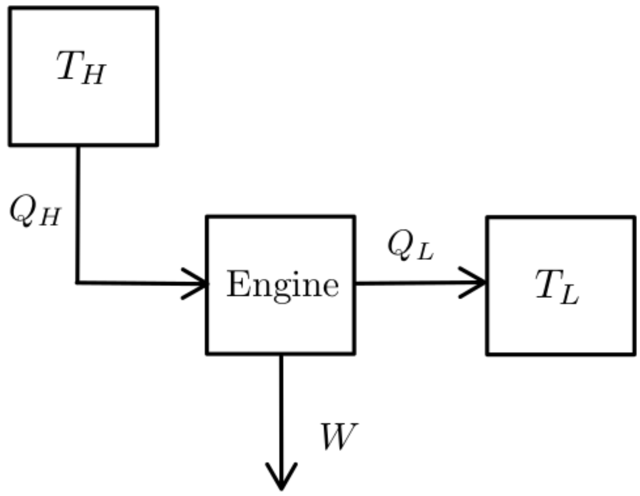


Figure 3.6: Simple schematic of a heat engine

The engine receives heat energy Q_H from hot reservoir T_H, converts it into useful work W, and some of energy is lost to the cold reservoir T_L. We define a new quantity:

Definition 3.6.2 (Efficiency of Heat Engine)**.**

$$\varepsilon = \frac{|W|}{|Q_H|} = \frac{|Q_H| - |Q_L|}{|Q_H|}$$

A version of the second law applied to heat engines is:

It is impossible to create an ideal heat engine, a type of engine that has $\varepsilon = 100\%$*.*

Definition 3.6.3 (Refrigerator)**.** *A refrigerator is a device that uses work in order to transfer energy from a low-temperature reservoir to a high-temperature reservoir.*

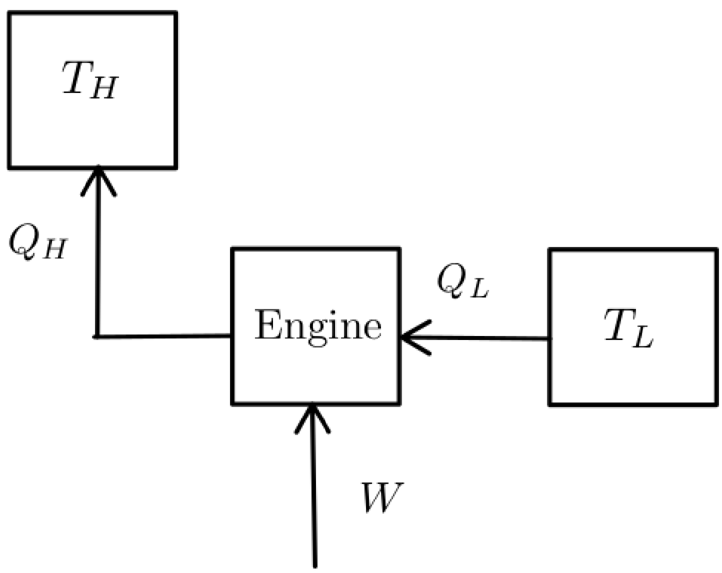


Figure 3.7: Simple schematic of a refrigerator

This engine receives work W to transfer heat Q_L from cold reservoir T_L to hot reservoir T_H. We define a new quantity:

Definition 3.6.4 (Coefficient of Peformance)**.**

$$K = \frac{|Q_L|}{|W|}$$

A version of the second law applied to refrigerators is:

It is impossible to create an ideal refrigerator, a type of engine that has $K = +\infty$*.*

The Carnot cycle is the simplest thermodynamics process capable of producing work. For simplicity, I will only consider the Carnot cycle for heat engine (for refrigerator, the cycle proceeds in the reverse direction).

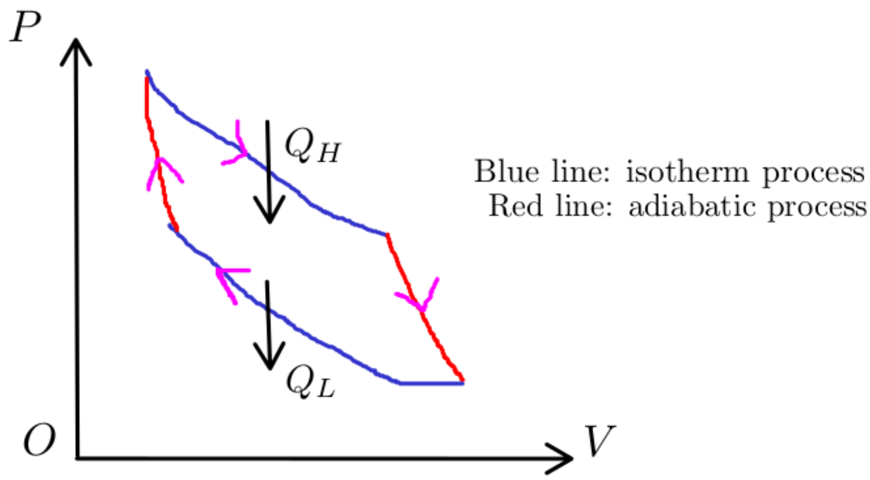


Figure 3.8: A very simple Carnot cycle

A very interesting consequence of the second law is that, since a perfect Carnot engine can not exist, the Carnot cycle is, in essence, **irreversible**[1].

[1]Since this is not our main topic, I briefly introduce some important concepts here. For more details (or explanations) regarding the relationship between entropy and engines, you should refer to sections 20-2, 20-3, Halliday textbook; or chapter 4, "An Introduction to Thermal Physics", Daniel V. Schroeder.

Chapter 4

Enthalpy and Free Energies

If you drop an aluminum ball into a drain cleaner solution; it is already in **constant pressure condition**:

$$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$$[1]

Now if you burn hydrogen gas inside a tube filled chlorine gas, you will observe the light blue flame, and hydrochloric gas is created:

$$H_2 + Cl_2 \rightarrow 2HCl$$[2]

This experiment is conducted in **constant volume condition**. When conducting any chemical experiment, we always concern with the operating conditions, either at constant volume or pressure. In this chapter, we study the influence of energy in chemical reactions.

4.1 Enthalpy

A cup of water is boiling on the stove; you can see the steam rising, and you might wonder how much energy the stove supplied to create that steam?

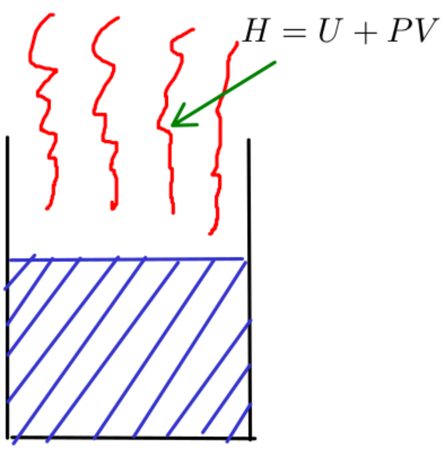


Figure 4.1: How much energy required to create steam from out of nothing?

[1] You can see the experiment here.
[2] You can see the experiment here.

The answer is quite simple: first you have to create its energy (U), then you add some work needed to make space for it (PV); so the total energy required is $U + PV$. Somehow if you want to annihilate the steam; you must destroy not only its internal energy, but also the work used to make its space.

Definition 4.1.1 (Enthalpy Definition)**.** *Enthalpy is the sum of a system's internal energy and the product of its pressure and volume:*

$$H = U + PV \quad (J)$$

Like entropy, we actually do not care much about the enthalpy H of a stable system; it is not so meaningful when I say "A cup of water has $H = \cdots$ (J)". The power of this quantity is most clearly demonstrated when we work with chemical reactions (or any transformation process). Let me give you an intuitive example:

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{H_2O} \quad (\Delta H = -286\mathrm{kJ/mol})$$

I have always thought of enthalpy as a representation of the state of all substances, in terms of energy. Look on the left side of reaction, to produce a mole of H_2 and a half mole of O_2, we need H_i joules. When the reaction occurs, H_i loses 286kJ to the surrounding environment to produce H_f, which contains only one mole of H_2O. Now I can conclude that because 286kJ is **released** to the surrounding environment, our reaction is **exothermic**.

$$\Delta H = H_f - H_i = -286\mathrm{kJ}$$

Conversely, the surrounding environment needs to **provide** energy to decompose H_2O into H_2 and O_2; this is called an **endothermic** reaction.

$$\mathrm{H_2O} \rightarrow \mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \quad (\Delta H = +286\mathrm{kJ/mol})$$

Example 4.1.1. The enthalpy of vaporization of water at 100°C is 40.63 (kJ/mol). How much energy is needed to create space for steam?
From Definition 4.1.1, we see that:

$$H = U + PV$$

The amount of work PV can be caculated by using ideal gas law:

$$PV = Nk_BT = nRT = RT = 3.10\ (\mathrm{kJ})$$

Under **constant pressure** condition, by performing some mathematical transformations, we obtain:

$$\begin{aligned}\Delta H = \Delta U + P\Delta V &= (Q + W) + P\Delta V = (Q + W_{\text{other}} - P\Delta V) + P\Delta V \\ &= Q + W_{\text{other}}\end{aligned}$$

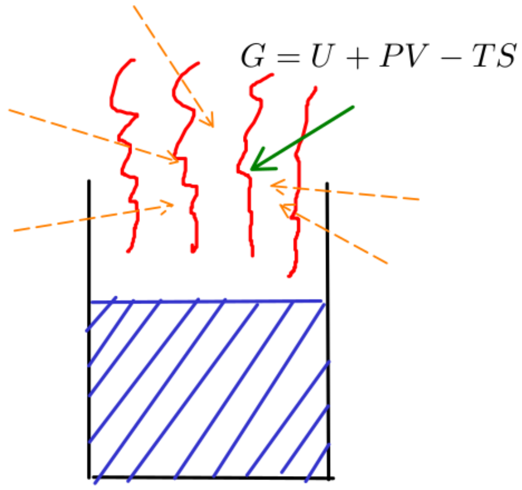
Have you notice a new quantity W_{other}? Unlike us, chemists have many ways to supply work to system without affecting its volume or pressure, and one of their most common methods is **electrolysis**. A classical example of electrolysis experiment is decomposing water to hydrogen and oxygen gases. If we do not provide any other types of work rather than W, we can redefine the quantity C_P as follows:

$$C_P = \frac{Q}{\Delta T} = \left(\frac{\partial H}{\partial T}\right)_P$$

4.2 Gibbs Free Energy

If you take into account the influence of the surrounding environment on the steam rising experiment, you will realize that this process also receives a small amount of "free energy" from the environment, so the total energy we have to provide is smaller than H:

$$G = H - TS = U + PV - TS$$



Free energy TS from environment enters system

Figure 4.2: Definition of Gibbs free energy

Definition 4.2.1 (Gibbs Free Energy Definition). *The Gibbs free energy is defined as:*

$$G = H - TS = U + PV - TS \quad (J)$$

Like enthalpy H, we do not really care about the total Gibbs free energy G of a system rather than the difference ΔG for any thermodynamics processes. The quantity ΔG is very useful for chemists to determine if a chemical reaction is **spontaneous** or not under **constant pressure condition**:

$$\Delta G = \Delta H - T\Delta S$$

We can find the values of H and S of any substances in the table "Thermodynamics Properties" at 298K and 1 bar. But how does it work? Basically, the logic is quite simple:

- If $\Delta H < 0$ and $\Delta S > 0$, that means $\Delta G < 0$, the reaction is always spontaneous at all temperature.
- If $\Delta H < 0$ and $\Delta S < 0$, that means the reaction is only spontaneous at low temperature.
- If $\Delta H > 0$ and $\Delta S < 0$, that means $\Delta G > 0$, the reaction is never spontaneous at all temperature.
- If $\Delta H > 0$ and $\Delta S > 0$, that means the reaction is only spontaneous at high temperature.

Let's look carefully on our quantities. From definitions above, we already know that both ΔH and ΔG indicate how much energy we have to add on our system. If they are negative, the reaction naturally wants to break chemical bonds inside, and releases its energy to the surrounding. The difference ΔS tells us the tendency of a relation to become chaotic. According to the second law, **any spontaneous process** tends to become more **chaotic** (or disordered); therefore, if $\Delta S < 0$, it means we must supply some external energy to "reverse" the reaction from chaotic to orderly.

Example 4.2.1. Determine whether this chemical reaction is spontaneous or not:

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{H_2O}$$

Everyone knows that $\Delta S < 0$ (water is more "ordered" than gases), so can we conclude this reaction is not spontaneous? After searching for values in the table, I found some numerical values: $H_{\mathrm{H_2}} = H_{\mathrm{O_2}} = 0$ (kJ), $H_{\mathrm{H_2O}} = -241.82$ (kJ); $S_{\mathrm{H_2}} = 130.68$ (J/K), $S_{\mathrm{O_2}} = 205.14$ (J/K) and $S_{\mathrm{H_2O}} = 188.83$ (J/K). The Gibbs free energy is:

$$\Delta G = \Delta H - T\Delta S = 1000.(-241.82) - 298.\left(188.83 - \frac{1}{2}.205.14 - 130.68\right) = -228.58 \quad (\mathrm{kJ})$$

Wow, so even $\Delta S < 0$, this reaction is still **spontaneous** at normal room condition, because $\Delta G < 0$. Observing the process, we can see that:

$$\Delta H = -241.82 \quad (\mathrm{kJ})$$
$$T\Delta S = -13.23 \quad (\mathrm{kJ})$$

The amount of energy needed for decreasing entropy is just 13.23kJ, but the reaction releases 241.82kJ to surrounding; so we still do not need to supply any energy here, the reaction is already **spontaneous**.

4.3 Helmholtz Free Energy

Enthalpy H and Gibbs free energy G are very powerful tools when we work with experiments at constant pressure condition. With that same line of thinking, at **constant volume condition**, the total energy we have to provide to create a system from nothing is:

$$F = U - TS$$

Definition 4.3.1 (Helmholtz Free Energy Definition)**.** *The Helmholtz free energy is defined as:*

$$F = U - TS \quad (J)$$

We do not have much to discuss Helmholtz energy here (until the next chapter), because in fact, it is much easier to conduct our experiment at constant pressure condition (yes, we all live in the same atmosphere, under the same pressure); so, to conclude this section, I will introduce a simple "mantra" that I use to remember the relationship between the 4 energy quantities:

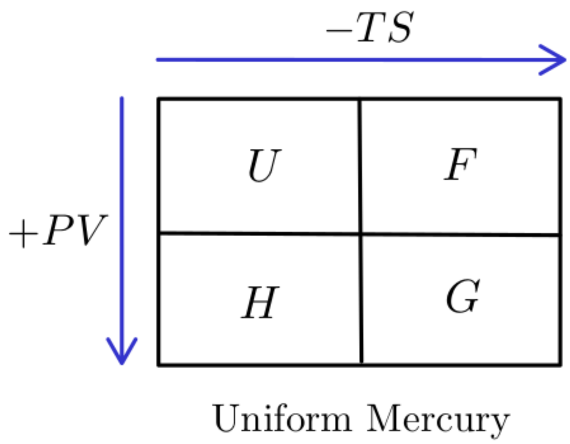


Figure 4.3: "Uniform Mercury"

4.4 Legendre Transformation

By applying the thermodynamics identity (Theorem 3.5.1), we obtain several useful results as follows:

$$\begin{aligned} dH &= dU + PdV + VdP \\ &= (TdS - PdV + \mu dN) + PdV + VdP \\ &= TdS + \mu dN + VdP \end{aligned}$$

This is an example of **Legendre transformation**; there are many ways you can do transformation between thermodynamics quantities, but here I only care about the two most important results:

$$\begin{aligned} dF &= dU - TdS - SdT \\ &= (TdS - PdV + \mu dN) - TdS - SdT \\ &= -PdV + \mu dN - SdT \end{aligned}$$

$$\Rightarrow S = -\left(\frac{\partial F}{\partial T}\right)_{N,V}$$

Similarly, for Gibbs free energy:

$$dG = dF + PdV + VdP = \mu dN - SdT + VdP$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_{N,P}$$

These 2 relationships are so important, they verify our intuition is correct: we have to provide more external energy to decrease the system entropy, and vice versa.

Chapter 5

Boltzmann Statistics

5.1 Boltzmann Distribution

In this chapter, we will learn how to apply some rules of probability and statistics to calculate the probability that a particle is in any given microstate. For example, inside a gas container, we can calculate the probability of finding particles moving with particular speed range, or any other states. Firstly, let's derive the probability density function:

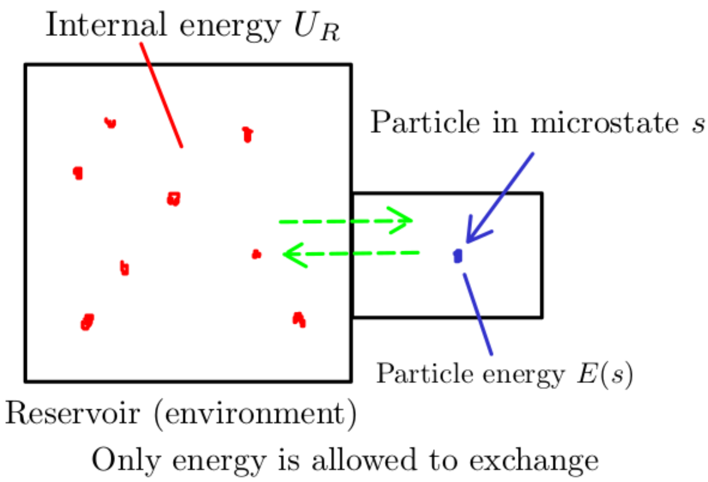


Figure 5.1: An imaginary experiment

Consider a system divided into two parts. The first part is the single particle found in a microstate s, and the second part is the rest of the container (or what I call the reservoir or environment). For any microstates s_1 and s_2, we have:

$$\frac{\mathcal{P}(s_1)}{\mathcal{P}(s_2)} = \frac{\Omega_R(s_1)}{\Omega_R(s_2)} = \exp\left(\frac{S_R(s_1) - S_R(s_2)}{k_B}\right)$$

From the thermodynamics identity (Theorem 3.5.1):

$$dS = \frac{dU}{T} \quad (V \text{ and } N \text{ are held as constants})$$

For abbreviation, I denote:

$$\beta = \frac{1}{k_B T}$$

We obtain:

$$\begin{aligned}
\frac{\mathcal{P}(s_1)}{\mathcal{P}(s_2)} &= \exp\left(\beta(U_R(s_1) - U_R(s_2))\right) \\
&= \exp(-\beta(E(s_1) - E(s_2))) \\
\Leftrightarrow \frac{\mathcal{P}(s_1)}{\mathcal{P}(s_2)} &= \frac{\exp\left(-\beta E(s_1)\right)}{\exp(-\beta E(s_2))}
\end{aligned}$$

Finally for any microstate s, we always have the proportional relationship:

$$\mathcal{P}(s) \propto e^{-\beta E(s)}$$

The probability density $\mathcal{P}(s)$ is formed by normalizing function Z as follows:

$$Z = \sum_s e^{-\beta E(s)} \Rightarrow \mathcal{P}(s) = \frac{1}{Z} e^{-\beta E(s)}$$

Z is the **partition function**, and exponential factor $e^{-\beta E(s)}$ is called the **Boltzmann factor**.

Definition 5.1.1 (Boltzmann Distribution)**.** *The probability of finding particle in any microstate s is:*

$$\mathcal{P}(s) = \frac{1}{Z} e^{-\beta E(s)}$$

*with Z is the **partition function**:*

$$Z = \sum_s e^{-\beta E(s)}$$

Corollary 5.1.0.1 (Average Energy)**.** *The average energy of any system is:*

$$\overline{E} = -\frac{\partial \ln Z}{\partial \beta}$$

Proof.

$$\overline{E} = \sum_s E(s)\mathcal{P}(s) = \frac{1}{Z}\sum_s E(s) e^{-\beta E(s)} = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}$$

□

This above result can be used to derive the equipartition theorem (Theorem 1.5.1); for any quadratic degree of freedom, its energy has the form:

$$E(q) = cq^2$$

The partition function Z is:

$$\begin{aligned}
Z &= \sum_s e^{-\beta E(s)} = \sum_s e^{-\beta c q^2} = \frac{1}{\Delta q}\int_{-\infty}^{+\infty} e^{-\beta c q^2} dq = \frac{1}{\Delta q \sqrt{\beta c}} \int_{-\infty}^{+\infty} e^{-q^2} dq \\
&= \frac{1}{\Delta q}\sqrt{\frac{\pi}{\beta c}}
\end{aligned}$$

We can see that:

$$\overline{E} = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = \frac{1}{2}\beta^{-1} = \frac{1}{2} k_B T$$

Our obtained result once again, verify the correctness of equipartition theorem.

5.2 Maxwell Speed Distribution

What is the probability of finding particle moving at speed range (v_1, v_2)? Well, the first thing we need to do is deriving a probability density function of speed $\mathcal{D}(v)$. But here is the subtle observation, a particle moving at speed v has an infinite velocity vector $\vec{v}$ representing how it moves in 3D space. All these velocity vectors form a "velocity sphere", which is similar to the concept of momentum space (as we discussed in Section 2.3):

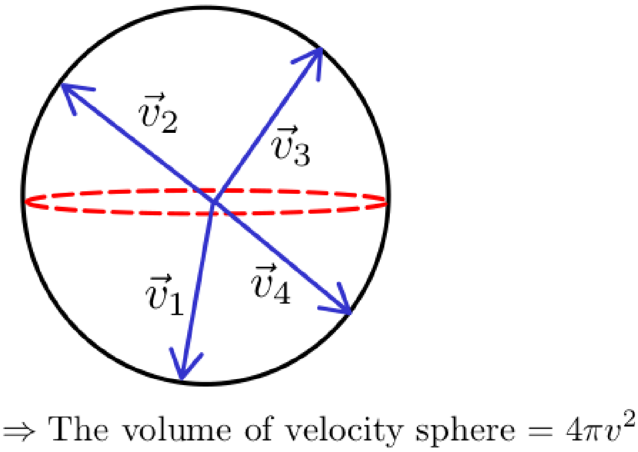


Figure 5.2: The velocity sphere and its volume

The probability of finding any particle moving at speed $\vec{v}$ is:

$$P(\vec{v}) \propto \exp\left(-\beta.\frac{1}{2}mv^2\right)$$

We are ready to plug every piece and build the $\mathcal{D}(v)$ function:

$$\mathcal{D}(v) \propto \exp\left(-\beta.\frac{1}{2}mv^2\right).4\pi v^2$$

The probability density function $\mathcal{D}(v)$ is rewritten as:

$$\mathcal{D}(v) = C.\exp\left(-\beta.\frac{1}{2}mv^2\right).4\pi v^2$$

Our final step is determining the constant C:

$$\begin{aligned}
1 &= C.\int_0^{+\infty} \exp\left(-\beta.\frac{1}{2}mv^2\right).4\pi v^2 dv \\
\Rightarrow \frac{1}{C} &= \int_0^{+\infty} \exp\left(-\beta.\frac{1}{2}mv^2\right)4\pi v^2 dv \\
&= 4\pi\left(\beta.\frac{1}{2}m\right)^{-3/2}\int_0^{+\infty} \exp\left[-\left(v\sqrt{\beta.\frac{1}{2}m}\right)^2\right]\left(v\sqrt{\beta.\frac{1}{2}m}\right)^2 .d\left(v\sqrt{\beta.\frac{1}{2}m}\right) \\
&= 4\pi\left(\beta.\frac{1}{2}m\right)^{-3/2}\int_0^{+\infty} e^{-v^2}v^2 dv
\end{aligned}$$

Integrating by part yields:

$$\int_0^{+\infty} e^{-v^2} v^2 dv = -\frac{1}{2}.\int_0^{+\infty} v.d(e^{-v^2}) = \frac{1}{2}\int_0^{+\infty} e^{-v^2} dv = \frac{\sqrt{\pi}}{4}$$

We obtain:

$$C = \left(\frac{\beta.m}{2\pi}\right)^{3/2}$$

Definition 5.2.1 (Maxwell Speed Distribution)**.** *The probability density function of finding any particle at a particular speed range is:*

$$\mathcal{D}(v) = \left(\frac{\beta.m}{2\pi}\right)^{3/2}.\exp\left(\frac{-\beta m v^2}{2}\right).4\pi v^2$$

Chapter 6

Magnetism in Matter

I think I can safely say that nobody understands quantum mechanics.

Richard Feynman

6.1 Magnetic Dipoles and Diamagnetism

6.1.1 Magnetic Dipoles

I think this final chapter is the most subtle in the book. From the first chapter until now, we have only been working with the assumption that all matter is made up of randomly moving "particles" without considering their structure. Do you remember Atomic Theory from the previous chapter (Theorem 1.1.1)? We treat it as the foundation (or as I said, an axiom). But we are having trouble explaining the essence of magnetism, our old theory, simply reaches its own limit. Now we must use a bit of the Bohr's model to keep everything on track. Let's begin with a simple experiment, imagine you have a single loop of wire between two magnetic poles. Inside the loop, a current flows counterclockwise. Applying left-hand rule, you can draw the direction of Lorentz force, which causes the loop to rotate:

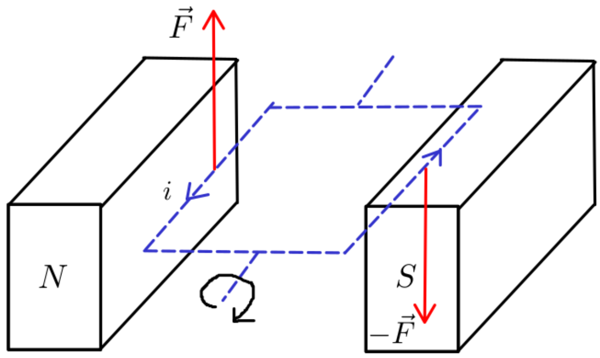


Figure 6.1: A loop inside magnetic field $\vec{B}$

A rectangular loop has size $a \times b$, under the net torque:

$$\vec{\tau}_{\text{net}} = \sum \vec{\tau} = \sum \vec{R} \times \vec{F} = \sum \vec{R} \times i(\vec{L} \times \vec{B})$$

I define $\vec{n}$ as the normal vector of the loop, and an angle $\theta = (\vec{n}, \vec{B})$, because both torques vectors are in the same direction, we obtain:

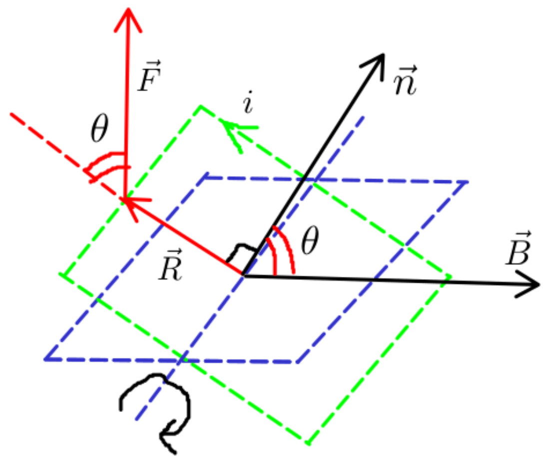


Figure 6.2: Both torque vectors in pointing out from the plane of paper

The scalar τ_{net} value is:

$$\tau_{\text{net}} = 2.\frac{b}{2}.iaB\sin\theta = iab\sin\theta.B = iA\sin\theta.B$$

I define a new vector $\vec{\mu}$ quantity, called **magnetic dipole moment**, same direction to normal vector $\vec{n}$, and has magnitude:

$$|\vec{\mu}| = iA = iab$$

Rewriting the net torque equation yields:

$$\vec{\tau}_{\text{net}} = \vec{\mu} \times \vec{B}$$

Definition 6.1.1 (Magnetic Moment Definition)**.** *The **magnetic moment** (or magnetic dipole moment) is a vector quantity that determines the magnitude of torque in a given magnetic field:*

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

Each angle θ (position), it is characterized by a **potential energy value**. I randomly choose 2 angles θ_i and θ_f:

$$W = -\Delta U = \int_{\theta_i}^{\theta_f} \tau d\theta = -|\vec{\mu}|.|\vec{B}|.(\cos\theta_f - \cos\theta_i) = -(U_f - U_i)$$

I assign the potential energy value to angle θ:

$$U(\theta) = -\vec{\mu}\cdot\vec{B}$$

From here we have a very important result: a magnetic dipole has its lowest energy when its vector $\vec{\mu}$ is lined up with magnetic field $\vec{B}$, and vice versa.

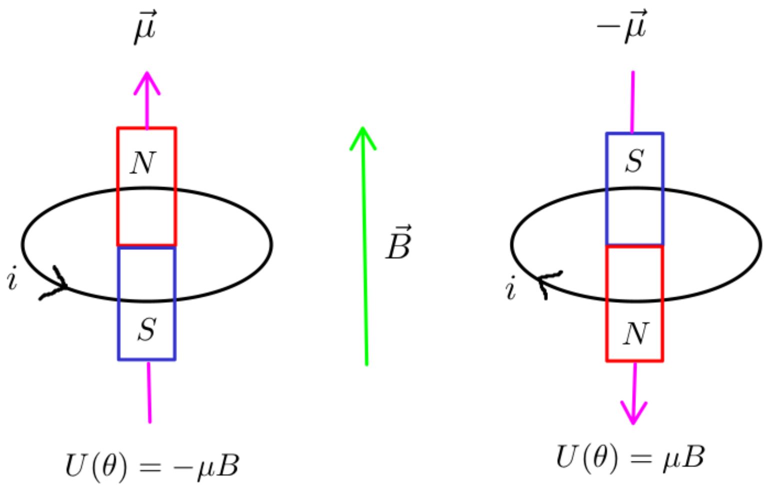
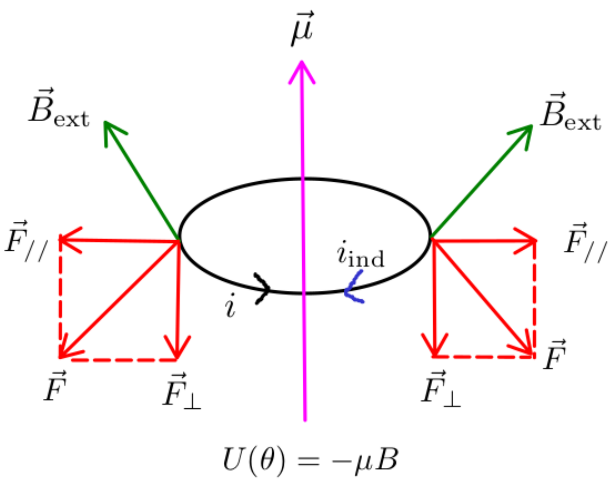


Figure 6.3: Highest and lowest potential energy value

To flip an arrow (magnetic moment vector) down, we must supply $2\mu B$ external energy units.

6.1.2 Diamagnetism

Before stepping in diamagnetism, let's perform a cool experiment first. Instead of uniform magnetic field $\vec{B}$, now I use a diverging source and its intensity gradually **increases**.



$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Due to Lenz's law, current i slowly decreases

$\vec{F}_{\text{net}} = \sum \vec{F} = \sum \vec{F}_{//} + \sum \vec{F}_{\perp} = \sum \vec{F}_{\perp}$

This current loop is **pulled down**.

Figure 6.4: A counterclockwise current inside a diverging magnetic field

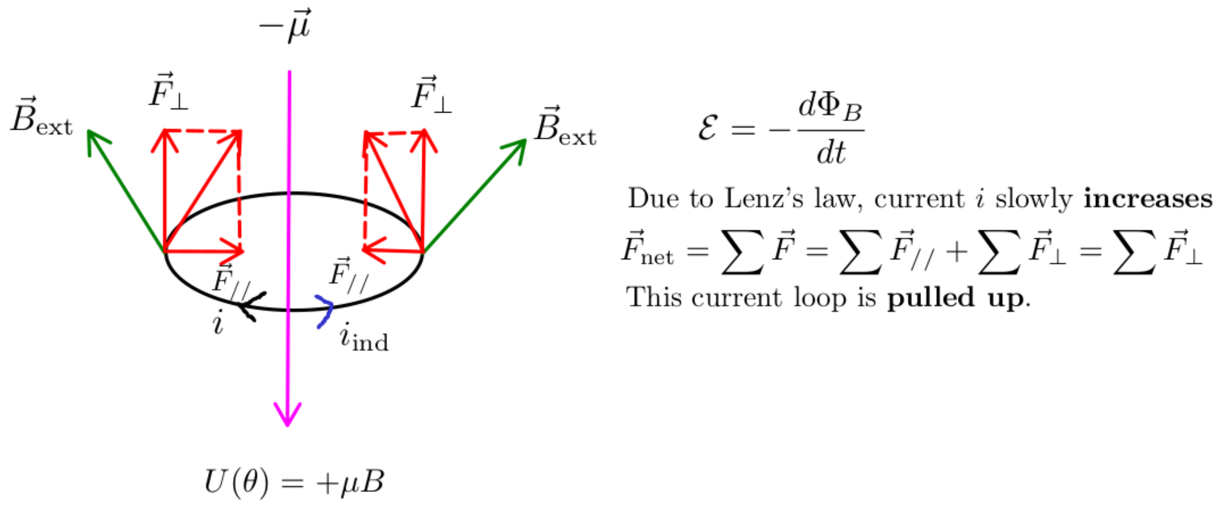


Figure 6.5: A clockwise current inside a diverging magnetic field

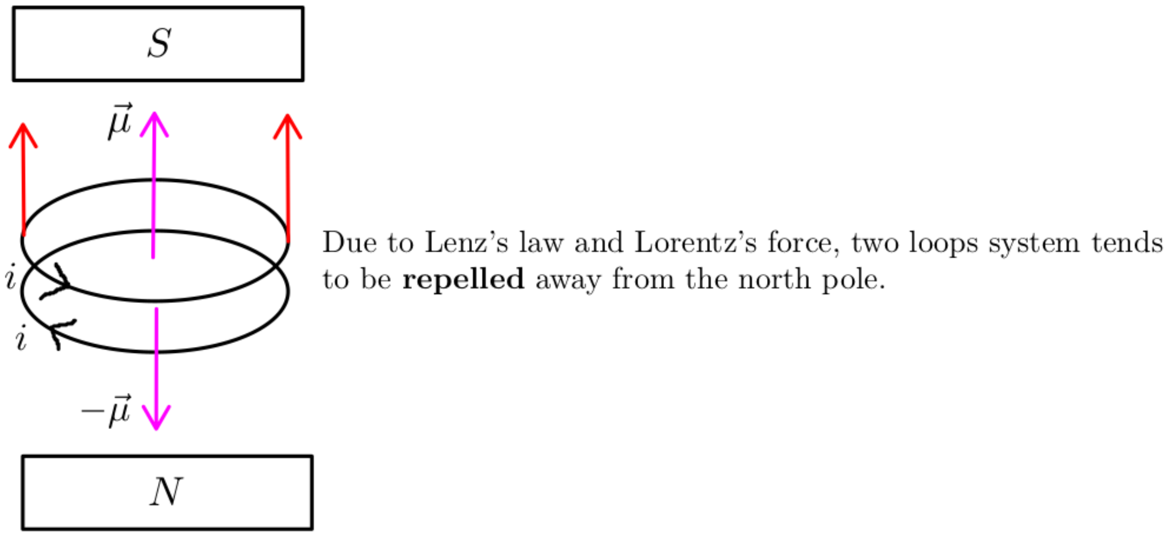


Figure 6.6: Two loops system tends to be **repelled away** from the north pole

Inside Bohr's atomic model, each electron has its own unique quantity called **spin**. Spin determines the direction of an electron's motion (clockwise or counterclockwise). If a material has perfectly matched "loops"[1] (electron orbits around nucleus), it tends to be **repelled from stronger magnetic field**; and we call such a material **diamagnetic**[2].

[1]Chemistry textbooks only go as far as explaining how molecules exhibit (or lack) diamagnetic property by drawing **molecules orbital diagram**; and now we fully understand why "paired" electrons can cause diamagnetism.

[2]Most biological bodies are diamagnetic materials, meaning that we can make them float in mid-air using a sufficiently strong magnetic field: Frog floats in mid-air experiment.

6.2 Paramagnetism

6.2.1 The Macroscopic View of Paramagnetism

If a material does not have any matched "loops" (or "unpaired" electrons), it tends to be **attracted from stronger magnetic field**, thanks to the effect of magnetic moment $\vec{\mu}$[1].

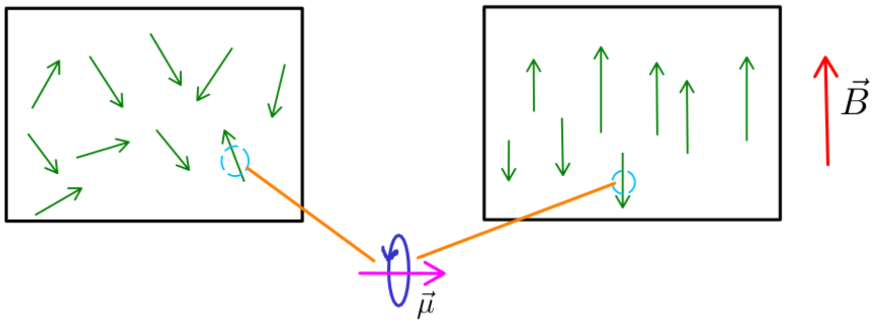


Figure 6.7: The macroscopic view of paramagnetic material

An external magnetic field $\vec{B}$ immediately magnetizes the material, forcing the random vectors to align; but when the magnetic field is removed, the material returns to its original state. We define the concept of **magnetization** of a sample, which represents the density of magnetizing dipole moment per unit volume.

Definition 6.2.1 (Magnetization Definition)**.**

$$M = \frac{measured\ magnetic\ moment}{V}$$

In 1985, Pierre Curie discovered experimentally the relationship:

$$M \propto \frac{B}{T}$$

This proportional relationship seems like very obvious; the higher the temperature, the harder the alignment of magnetic moment vectors, and vice versa.

6.2.2 The Microscopic View of Paramagnetism

To examine the paramagnetic material, and explain some surprising phenomenons, we have to equip statistical tool. For simplicity, I restrict our concern to **two-state paramagnet**. Two-state paramagnet are the materials in which the magnetic moment vector can only point either up or down.

[1]A very well-known example of a paramagnetic material is liquid oxygen, in contrast to liquid nitrogen, a diamagnetic material. You should draw their molecules orbital diagram to see the difference.

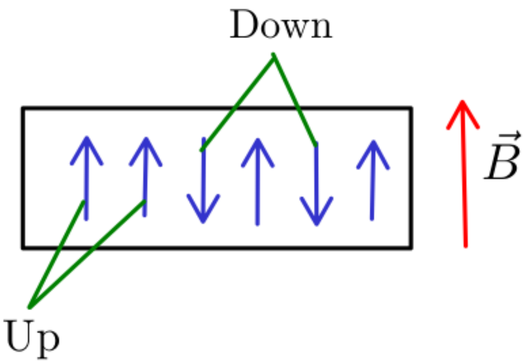


Figure 6.8: Two-state paramagnet

If I denote N as "the number of", the multiplicity and entropy of the system are:

$$\Omega(N_\uparrow) = \binom{N}{N_\uparrow} \Rightarrow S = k_B \ln \Omega = k_B \ln \binom{N}{N_\uparrow}$$

The internal energy is determined by:

$$U = \mu B(N_\downarrow - N_\uparrow) = \mu B(N - 2N_\uparrow)$$

I neglect the constant B in the magnetization formula, and redefine the M quantity as:

$$M = \mu(N_\uparrow - N_\downarrow) = -\frac{U}{B}$$

Before we look for an analytical solution, let's consider the visual representation for these things. We all know how binomial function work, so I can draw a simple table with 4 columns:

$N_\uparrow$	S	U	M
0	0	(+) max	(−) min
$N/2$	(+) max	0	0
N	0	(−) min	(+) max

Blue arrow: increasing direction
Red arrow: decreasing direction

Figure 6.9: Relationship between $N_\updownarrow, S, U, M$

If our two-state paramagnet starts moving from state $N_\uparrow = 0$ to $N_\uparrow = N$, I can plot the graph of (U, S):

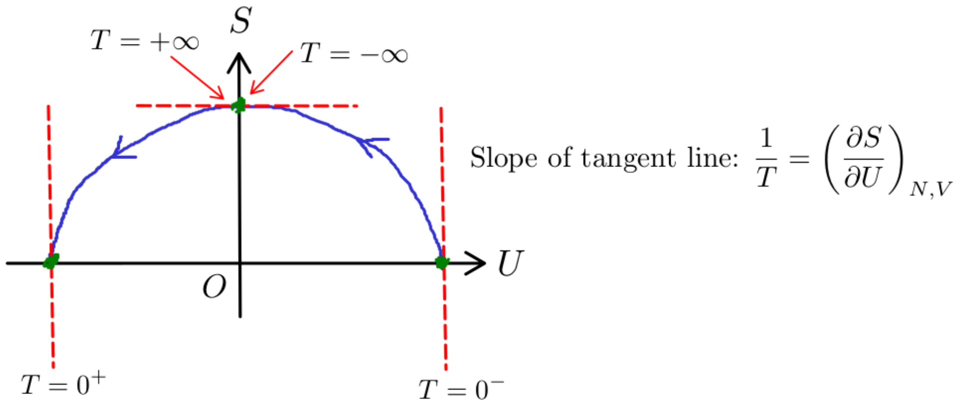


Figure 6.10: The graph of (U, S)

What can we see here? Recall the definition of temperature (Definition 3.2.2), we determine the temperature of our system at 4 special labeled points. Wow, this is the first time we observe the **negative temperature** appears; and seems like these states $U \to 0^+$ and $U \to 0^-$ are very unstable, corresponds to $N \to (N/2)^-$ and $N \to (N/2)^+$. Maybe the "exact state", where S is exactly equal to zero, can not happen.